

Slide 1 — Title

Sparse-Measurement Image Reconstruction

Your slide contains your project title, M.Tech details, your name and guide.

Say:

"Good morning everyone.

I am Aravind Ashok, M.Tech Computer Science and Engineering student.

Today I am going to present the current progress of my project titled **Sparse-Measurement Image Reconstruction**.

My project mainly focuses on reconstructing an image from a smaller number of measurements using a GAN-based approach. I have also studied conventional reconstruction methods for comparison."

Then move to the next slide.

Slide 2 — Contents

Your contents include Introduction, Problem Statement, Literature/Reference Papers, Initial Learning, GAN, Software, Dataset and Experimental Results.

Say:

"First, I will introduce the problem and explain the motivation of the project.

Then I will discuss the reference papers I studied and my initial learning about DNN and GAN.

After that, I will explain the software, dataset and experimental setup.

Finally, I will discuss the current reconstruction results, comparison and future work."

Don't spend much time here.

Slide 3 — Introduction

Your slide explains that the objective is to reconstruct an image from fewer measurements and that GAN is your main approach, with LASSO and SBL used for comparison.

Say:

"Image reconstruction means recovering an image when complete information about the original image is not directly available.

In my project, instead of directly using all the pixel information, I convert the image into a smaller number of measurements.

Then I use a reconstruction model to recover the original image.

My main focus is on GAN-based reconstruction. I am also using conventional methods such as LASSO and SBL for comparison."

If they ask "What is the basic idea?"

Point to your workflow and say:

"The basic idea is **original image, measurement generation, reconstruction, and finally comparison with the original image.**"

Slide 4 — Problem Statement

Your current slide uses **784 → 100 measurements**. It says MNIST is 28×28, giving 784 pixels, and the current measurement dimension is 100.

Say:

"An MNIST image has a size of 28 by 28 pixels, so there are 784 pixel values.

The problem I am studying is whether I can reconstruct this 784-dimensional image when I have only a smaller number of measurements.

In my current experiment, I use 100 measurements instead of the original 784 pixel values.

The mathematical model is $y = Ax$.

Here, x is the original image, A is the measurement matrix, and y is the compressed measurement."

Point to the diagram/equation while saying this.

If asked: "Why is this difficult?"

Say:

"Because the number of measurements is smaller than the number of unknown pixel values. Therefore, the reconstruction problem is underdetermined."

Slide 5 — Literature / Reference Paper

Your first paper is **Sparse Signal Recovery Using Deep Neural Networks**, by Rubin Jose Peter and Chandra R. Murthy. It is identified in your PPT as your main paper.

Say:

"Before implementing my model, I studied existing research related to sparse signal recovery and image reconstruction.

My main reference paper is **Sparse Signal Recovery Using Deep Neural Networks**, by Rubin Jose Peter and Chandra R. Murthy.

The main concept I learned from this paper is how a signal can be recovered from a smaller number of measurements.

This paper gave me the basic foundation for understanding the measurement and reconstruction problem."

Then:

"I also studied additional papers related to GAN-based reconstruction and compressed sensing."

Slide 6 — Literature Review: Additional Papers

Your PPT contains the papers on single-pixel imaging, MRI reconstruction and lung CT reconstruction.

Say:

"The second paper studies GAN-based reconstruction for single-pixel imaging. The important concept I learned from this paper is how measurement information can be used to guide GAN reconstruction.

The third paper combines compressed sensing and GAN-based reconstruction for MRI using undersampled k-space data.

The fourth paper applies GAN and compressed sensing to ultra-sparse-view lung CT reconstruction.

So, from these papers, I understood that GAN-based reconstruction can be used when the available image information is incomplete or limited."

Then make the connection:

"However, these papers focus on specific applications such as single-pixel imaging, MRI and CT. My current work starts with MNIST as an initial experimental dataset."

If your guide asks "What is the gap?"

Say:

"The papers I studied address specific reconstruction applications. My current work focuses on establishing and evaluating a measurement-based GAN reconstruction framework on MNIST, followed by comparison with conventional sparse-recovery methods and future testing on additional datasets."

Slide 7 — Overall Reconstruction Pipeline

Your PPT has an overall reconstruction pipeline after the literature section.

This is an important slide. Say slowly:

"This is the overall workflow of my project.

First, I take the original MNIST image.

Since the image is 28 by 28, it contains 784 pixel values.

I flatten the image into a vector and apply a measurement matrix.

This produces a smaller measurement vector.

In my current experiment, I use 100 measurements.

These measurements are then given to the reconstruction model.

The model produces the reconstructed 784-pixel image.

Finally, I compare the reconstructed image with the original image using reconstruction metrics."

If the diagram shows GAN, point to the Generator and explain it briefly.

Slide 8 — Initial Learning Stage: DNN

Your slide says you initially studied DNN structure, neurons, weights, activation functions, forward propagation, loss and training.

Say:

"At the beginning of the project, I first studied the basic concepts of Deep Neural Networks.

I learned about the input, hidden and output layers, neurons, weights and activation functions.

I also studied forward propagation, loss functions and model training.

This helped me understand how a neural network learns a relationship between an input and an output.

This basic knowledge was then useful when I moved to GANs."

If asked:

"Why did you study DNN when your project is GAN?"

Say:

"GANs are also built using neural networks, so understanding the basic DNN concepts was necessary before studying GAN architecture and training."

Slide 9 — Learning About GAN

Your current PPT has a separate GAN-learning slide.

Say:

"After understanding the basic DNN concepts, I studied Generative Adversarial Networks.

A GAN consists mainly of two networks: the Generator and the Discriminator.

The Generator produces the reconstructed image, while the Discriminator tries to distinguish between real and generated images.

In my project, I use the compressed measurement as the input to the Generator.

Therefore, the Generator learns a mapping from the compressed measurements to the reconstructed image."

Very likely question:

"Are you giving random noise to your Generator?"

Say:

"No. In my reconstruction setup, the Generator is conditioned on the compressed measurement. The measurement vector is the main input used to reconstruct the image."

That's an important point.

Slide 10 — Software, Libraries and Functions

Your PPT contains a dedicated software/libraries slide.

Say:

"For implementation, I used Python as the main programming language.

NumPy is used for numerical operations and matrix calculations.

TensorFlow and Keras are used for building and training the GAN.

SciPy is used for mathematical operations such as DCT-related processing.

Scikit-learn is used for conventional methods such as LASSO.

Matplotlib is used for visualizing the reconstructed images and results."

If your slide contains specific functions, briefly mention the ones visible there rather than listing every Python function.

Slide 11 — Dataset and Experimental Setup

Your PPT uses MNIST, 28×28 images, 784 values and normalized pixel values.

Say:

"For my initial experiment, I used the MNIST handwritten digit dataset.

Each image is 28 by 28 pixels, so it contains 784 values.

The pixel values are normalized between 0 and 1.

I then generate a smaller measurement representation and use this as the input for reconstruction."

If asked "Why MNIST?"

"MNIST is a simple and well-defined dataset, so it is suitable for initially testing the reconstruction methodology. After establishing the approach, I plan to test additional datasets."

Slide 12 — Present: Conventional Comparison

Your PPT introduces LASSO and SBL here and explicitly states that they are comparison baselines rather than the main project.

Say:

"At the current stage, I have started comparing the GAN reconstruction with conventional sparse-recovery methods.

The two methods I have considered are LASSO and Sparse Bayesian Learning, or SBL.

LASSO is an optimization-based sparse-recovery method that uses L1 regularization.

SBL is a Bayesian approach to sparse recovery.

I am using both as conventional baselines. My main focus remains the GAN-based reconstruction."

If asked "Why LASSO?"

"LASSO gives me an optimization-based sparse-recovery baseline."

If asked "Why SBL?"

"SBL gives me a Bayesian sparse-recovery baseline, so I can compare the GAN with two different conventional approaches."

Don't explain all the LASSO/SBL mathematics here unless they specifically ask.

Slide 13 — Future Work

Your PPT currently places Future Work before the experimental results. It includes additional datasets, testing all models and studying different measurement dimensions.

Say:

"My next stage of work is to move beyond the MNIST dataset.

I plan to select additional image datasets and apply the same measurement and reconstruction framework.

I will test the GAN, LASSO and SBL models on these datasets.

I also plan to study the effect of different measurement dimensions and analyze how the reconstruction quality changes.

Based on these results, I can further improve the reconstruction model."

Slide 14 — Experimental Results: 784 → 100

Your PPT shows the original and reconstructed MNIST image and reports:

Normalized MSE = 0.12024844.

Say:

"Here I tested the reconstruction using 100 measurements from the original 784-dimensional image.

The normalized MSE obtained in this experiment is approximately 0.1202.

The reconstructed image shows that the model is able to recover the main structure of the original handwritten digit from a much smaller number of measurements."

Don't call it "excellent" unless you have a basis for that comparison.

Slide 15 — Experimental Results: 784 → 50

Your PPT shows **784 → 50** and normalized MSE of **0.30164862**.

Say:

"I also tested the model with an even smaller number of measurements.

Here the original 784 values are reduced to only 50 measurements.

The normalized MSE increased to approximately 0.3016.

This shows that when the number of available measurements is reduced further, the reconstruction becomes more difficult and the reconstruction error increases."

This is a very good point to emphasize.

Your guide may ask:

"What did you learn from this experiment?"

Say:

"I observed that the measurement dimension has a direct effect on reconstruction quality. With fewer measurements, less information is available for reconstruction, so the error increased."

Slide 16 — Performance Evaluation

Your slide contains MSE, NMSE and NMSE(dB).

Say:

"To quantitatively evaluate the reconstructed images, I use MSE, NMSE and NMSE in dB.

MSE measures the average squared difference between the original and reconstructed pixels.

NMSE normalizes this error with respect to the energy of the original image.

NMSE in dB represents the normalized error on a logarithmic scale.

These metrics are useful for comparing different reconstruction methods."

If asked "Which is better?"

For MSE/NMSE:

"A lower error indicates a smaller difference between the reconstructed and original image."

For NMSE(dB):

"A more negative NMSE(dB) corresponds to a smaller NMSE."

Slide 17 — References

Your references include your main paper and the supporting GAN/compressed-sensing papers.

Say:

"These are the main references I used during my literature study and implementation."

Don't read every reference word-for-word.

If your guide asks:

"Which is your main reference?"

Immediately answer:

"The first reference, *Sparse Signal Recovery Using Deep Neural Networks*, is my main reference paper."

Slide 18 — Thank You

Say:

"That concludes my current project progress. Thank you for listening. I am ready to take your questions."

Then stop.

The 10 questions I would prepare most carefully

For your particular PPT, I would prepare these first:

1. What exactly is your project?

"I am studying reconstruction of an image from a smaller number of measurements using a GAN-based reconstruction approach, with LASSO and SBL used as conventional comparison methods."

2. What is $y = Ax$?

"x is the original image, A is the measurement matrix, and y is the smaller measurement vector."

3. Why 784 → 100?

"MNIST has 784 pixels, and I want to study reconstruction when only a substantially smaller number of measurements are available."

4. Why did you reduce it to 50 also?

"To study how further reduction in measurements affects reconstruction quality."

5. Why did the error increase from 100 to 50?

"Because fewer measurements provide less information for reconstruction, making the problem more difficult."

Your actual results support this observation.

6. Why GAN?

"I wanted to investigate a learning-based reconstruction approach that can learn the mapping from compressed measurements to image reconstruction."

7. Why LASSO and SBL?

"They provide conventional sparse-recovery baselines for comparison with the GAN."

8. What is your contribution?

A careful answer:

"My current contribution is the implementation and experimental evaluation of a measurement-based GAN reconstruction framework, along with comparison against conventional sparse-recovery methods. The work is currently at the experimental stage, and I plan to extend it to additional datasets."

9. What have you completed?

"I studied the reference literature, learned DNN and GAN concepts, implemented the measurement and GAN reconstruction pipeline, tested it on MNIST, implemented LASSO and SBL comparison methods, and evaluated reconstruction using MSE and NMSE."

10. What will you do next?

"I will test the models on additional datasets, study different measurement dimensions, compare the reconstruction performance, and use the results to improve the model."

One correction I strongly recommend before you present

Your current PPT's **Contents slide does not explicitly mention the mathematical measurement model, comparison methodology, or future work as separate sections**, even though those topics appear later.

You can improve the contents to:

1. Introduction
2. Problem Statement
3. Project Objectives
4. Literature Review
5. Initial Learning — DNN
6. Learning About GAN
7. Proposed Reconstruction Pipeline
8. Software and Libraries
9. Dataset and Experimental Setup
10. Present Status — GAN, LASSO and SBL
11. Experimental Results
12. Performance Evaluation
13. Learning Outcomes
14. Future Work
15. Conclusion

That will make the presentation flow much more logically from **“what is the problem?”** → **“what did I study?”** → **“what did I implement?”** → **“what have I achieved?”** → **“what will I do next?”**.

ok give4

Yes. Here is the **final presentation speaking script**, slide by slide, in a form you can practice directly.

1. Title

“Good morning everyone. I am Aravind Ashok, M.Tech Computer Science and Engineering student. Today I am presenting the current progress of my project, **Sparse-Measurement Image Reconstruction**. My main focus is reconstructing images from a smaller number of measurements using a GAN-based approach.”

2. Contents

“First, I will introduce the problem and objectives. Then I will discuss the literature I studied and my initial learning of DNN and GAN. After that, I will explain the implementation, dataset and experimental results. Finally, I will discuss the current comparison and future work.”

3. Introduction

“Image reconstruction means recovering an image from incomplete or limited information. In my project, instead of directly using all the pixels, I generate a smaller number of measurements and use them to reconstruct the original image. My main approach is GAN-based reconstruction, while LASSO and SBL are used for comparison.”

4. Problem Statement

“An MNIST image has 28 by 28 pixels, which gives 784 pixel values. My objective is to reconstruct this image using fewer measurements. In the current experiment, I use 100 measurements instead of 784. The basic mathematical model is $y = Ax$, where x is the original image, A is the measurement matrix and y is the measurement vector.”

5. Objectives

“The main objectives are to understand sparse measurement and reconstruction, study DNN and GAN concepts, implement a GAN-based reconstruction model, compare it with conventional methods such as LASSO and SBL, and finally extend the experiment to additional datasets.”

6. Literature Review

“I started my literature study with the paper **Sparse Signal Recovery Using Deep Neural Networks** by Rubin Jose Peter and Chandra R. Murthy. This is my main reference paper. It helped me understand the basic problem of recovering a signal from a smaller number of measurements.”

“I then studied papers related to GAN-based reconstruction in single-pixel imaging, MRI and CT. These papers helped me understand how GANs and compressed sensing can be used when image information is incomplete.”

7. Overall Pipeline

“This is the overall workflow of my project. First, I take the original MNIST image. The 784 pixels are converted into a smaller number of measurements using a measurement matrix. These measurements are given to the reconstruction model, which generates the reconstructed 784-pixel image. Finally, I compare the reconstructed image with the original image using quantitative metrics.”

8. Initial Learning — DNN

“Initially, I studied the basic concepts of Deep Neural Networks. I learned about input, hidden and output layers, neurons, weights, activation functions, loss functions and training. This gave me the basic understanding needed to study GANs.”

9. Learning About GAN

“After learning DNN, I studied GANs. A GAN consists mainly of a Generator and a Discriminator. The Generator produces the reconstructed image, while the Discriminator distinguishes between real and generated images. In my project, the compressed measurement is used as the input to the Generator.”

10. Software and Libraries

“I used Python for implementation. NumPy was used for numerical and matrix operations, TensorFlow and Keras for the GAN, SciPy for mathematical operations, scikit-learn for LASSO, and Matplotlib for visualization.”

11. Dataset and Setup

“For the initial experiment, I used the MNIST handwritten digit dataset. Each image is 28 by 28 pixels, giving 784 values. The pixel values are normalized between 0 and 1. I then generate a lower-dimensional measurement representation for reconstruction.”

12. Present Status — Comparison

“At the current stage, I have implemented conventional comparison methods. LASSO is an optimization-based sparse-recovery method using L1 regularization, while SBL is a Bayesian sparse-recovery method. These methods are not the main focus of my project; they are used as comparison baselines for the GAN.”

13. Future Work

“My next work is to test the reconstruction framework using additional datasets apart from MNIST. I will test GAN, LASSO and SBL on these datasets and study how the reconstruction quality changes with different image types and measurement dimensions.”

14. Experimental Result — 784 → 100

“Here I tested the reconstruction using 100 measurements from the original 784-dimensional image. The normalized MSE obtained was approximately 0.1202. This shows that the model was able to reconstruct the basic structure of the image using significantly fewer measurements.”

15. Experimental Result — 784 → 50

“I then reduced the measurements further from 100 to 50. In this case, the normalized MSE increased to approximately 0.3016. This indicates that reconstruction becomes more difficult when fewer measurements are available.”

16. Performance Evaluation

“I use MSE, NMSE and NMSE in dB to evaluate the reconstruction. MSE measures the squared reconstruction error. NMSE normalizes the error with respect to the original image energy. NMSE in dB represents the normalized error on a logarithmic scale.”

17. Learning Outcomes

“Through this project, I learned the fundamentals of DNNs and GANs, sparse measurement and reconstruction, image preprocessing, implementation using Python and TensorFlow, conventional reconstruction methods, and evaluation using MSE and NMSE. I also gained practical experience in debugging and analyzing reconstruction results.”

18. References

“These are the main papers and references that I used for the literature study and implementation.”

19. Conclusion

“To conclude, I studied the problem of reconstructing an image from a smaller number of measurements. I studied DNN and GAN concepts, implemented a GAN-based reconstruction model using MNIST, and implemented LASSO and SBL as conventional comparison methods. The current results show that reducing the number of measurements from 100 to 50 increases the reconstruction error. My next step is to test the models on additional datasets.”

20. Thank You

“That concludes my project progress presentation. Thank you for listening. I am ready to take your questions.”

The one sentence to remember if you get nervous

If you forget what to say, remember your entire project as:

“I take a 784-pixel MNIST image, convert it into a smaller number of measurements, reconstruct the image using a GAN, compare it with conventional LASSO and SBL methods, evaluate the reconstruction using MSE and NMSE, and plan to test the framework on additional datasets.”

That single sentence captures your complete project.

Yes. For your project, think of **LASSO and SBL as two conventional reconstruction methods** that take the same compressed measurements that your GAN takes.

Your overall idea is:

Original image → measurement matrix → compressed measurements → reconstruction

For MNIST:

784 pixels → M measurements → 784 reconstructed pixels

For your current experiments, $M = 100$ or $M = 50$.

1. What is LASSO?

LASSO = Least Absolute Shrinkage and Selection Operator.

In your project, LASSO is used to recover the original image from fewer measurements by assuming that the image has a **sparse representation**.

Simple idea

Suppose the original image is:

$$x$$

You generate measurements:

$$y = Ax$$

where:

- x = original 784-pixel image
- A = measurement matrix
- y = compressed measurements

The problem is:

We know A and y , but we don't know x . How can we recover x ?

LASSO solves this optimization problem.

$$\hat{s} = \arg \min_s \left[\frac{1}{2} \|y - Bs\|_2^2 + \lambda \|s\|_1 \right]$$

where, in your DCT-based implementation,

$$B = AD$$

and

$$x = Ds$$

Here s is the sparse representation of the image.

2. Why do we use DCT with LASSO?

An image itself may not be very sparse.

But after converting the image into a suitable transform domain such as **DCT**, many coefficients can become small or close to zero.

So:

$$x \rightarrow DCT \rightarrow s$$

and we try to recover s .

Then:

$$s \rightarrow IDCT \rightarrow \hat{x}$$

So your LASSO pipeline is:

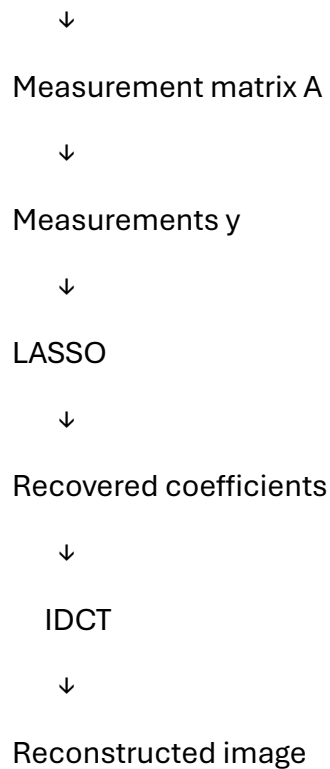
Original MNIST image

↓

DCT

↓

Sparse coefficients s



3. What does the L1 term do?

This part is important:

$$\lambda \|s\|_1$$

The L1 term encourages many coefficients to become **zero or very small**.

For example:

Before LASSO:

0.8 0.02 0.7 0.001 0.03 0.6 0.004 ...

After sparse recovery:

0.8 0 0.7 0 0 0.6 0 ...

That is why LASSO is useful for **sparse recovery**.

In one sentence for your presentation:

LASSO reconstructs the image by finding a sparse representation whose measurements match the observed measurements while using L1 regularization to promote sparsity.

4. How do you implement LASSO in your program?

Your program basically does these steps:

Step 1 — Read MNIST

$x = \text{image}$

Shape:

28×28

Flatten:

784

Step 2 — Apply DCT

$s = \text{dct}(x)$

Now you have the sparse representation.

Step 3 — Generate measurements

$y = A @ x$

But because you're using DCT:

$$y = A(Ds)$$

so:

$$y = (AD)s$$

Define:

$B = A @ D$

Then:

$y = B @ s$

Step 4 — LASSO solves for s

In Python, you can use:

```
from sklearn.linear_model import Lasso
```

```
model = Lasso(alpha=0.001, max_iter=1000)
```

```
model.fit(B, y)
```

```
s_hat = model.coef_
```

Step 5 — Convert back to image

```
x_hat = idct(s_hat)
```

Now `x_hat` is your reconstructed image.

5. What is SBL?

SBL = Sparse Bayesian Learning.

SBL also tries to recover a sparse signal from fewer measurements.

But its approach is different from LASSO.

LASSO

Uses:

L1 optimization

SBL

Uses:

Bayesian probability/statistical modeling

So:

Method Main idea

LASSO L1 optimization

SBL Bayesian sparse learning

GAN Neural-network-based reconstruction

6. Simple idea of SBL

Again we start with:

$$y = Bs$$

We don't know s .

SBL assumes that every coefficient of s has a probability distribution.

For example:

$$s_i \sim N(0, \gamma_i)$$

Here:

$$\gamma_i$$

represents how important that coefficient is.

If a coefficient is not useful:

$$\gamma_i \rightarrow 0$$

Therefore:

$$s_i \rightarrow 0$$

So SBL gradually identifies which coefficients are important.

7. How SBL works step-by-step

Suppose you have:

Measurements y

↓

Initial gamma values

↓

Calculate covariance

↓

Estimate coefficients

↓

Update gamma

↓

Repeat

↓

Final sparse coefficients

↓

IDCT

↓

Reconstructed image

The important idea is:

SBL repeatedly estimates the signal and updates the importance of each coefficient.

8. What happens inside your SBL code?

Your implementation has:

```
gamma = np.ones(n)
```

Initially, all coefficients are given equal importance.

Then:

```
C = B @ Gamma @ B.T + noise * I
```

In your simplified implementation, this is represented using:

```
BG = B * gamma[np.newaxis, :]
```

```
C = BG @ B.T + noise * np.eye(m)
```

Then the program estimates the coefficients:

```
mu = gamma * (B.T @ C_inv_y)
```

Then it calculates updated values:

```
gamma_new = mu**2 + Sigma
```

Then:

```
gamma = gamma_new
```

This is repeated.

For example:

Iteration 1

↓

estimate coefficients

↓

update gamma

Iteration 2

↓

estimate coefficients

↓

update gamma

Iteration 3

↓

estimate coefficients

↓

update gamma

...

Eventually, the algorithm produces:

$$\hat{s}$$

Then:

$$\hat{x} = D\hat{s}$$

or using IDCT:

`x_hat = idct(s_hat)`

9. Very important: Is SBL adding noise?

In your current code you have:

```
noise = 0.001
```

This **does not mean you are randomly adding noise to your MNIST image or measurement vector.**

You use it here:

```
C = BG @ B.T + noise * np.eye(m)
```

It provides a small diagonal term in the covariance calculation.

So when your guide asks:

"Are you adding noise?"

You can say:

"No. In my current implementation, I am not adding random noise to the measurement data. The 0.001 parameter is used as a small noise or regularization term inside the SBL covariance calculation."

10. LASSO vs SBL vs GAN

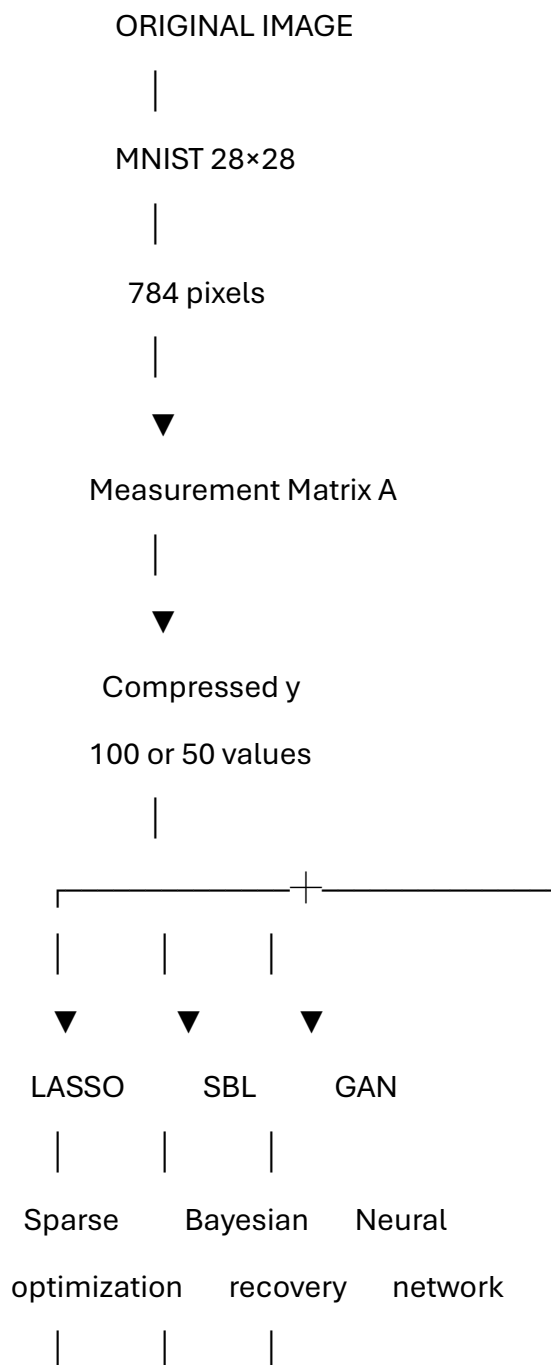
This is the most important comparison for your project.

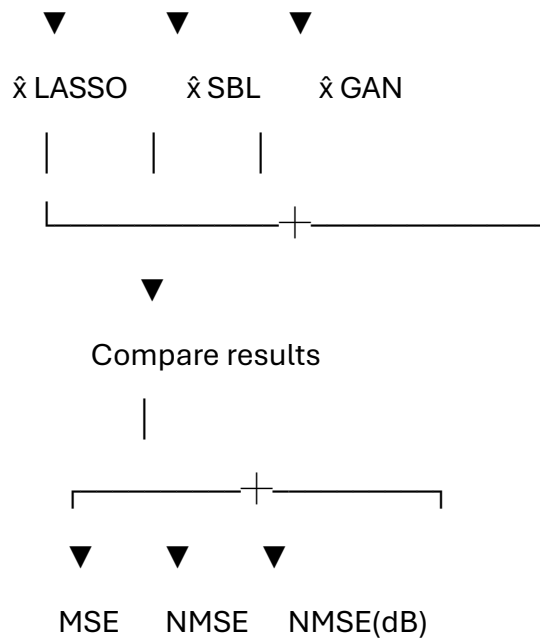
Feature	LASSO	SBL	GAN
Type	Optimization	Bayesian	Deep learning
Main principle	L1 sparsity	Bayesian sparsity	Generator + discriminator
Input	Measurements	Measurements	Measurements
Output	Reconstructed image	Reconstructed image	Reconstructed image
Iterative?	Yes, internally	Yes	Training epochs
Uses L1?	Yes	No	No
Uses probability model?	No	Yes	Not in the same sparse Bayesian sense

Feature	LASSO	SBL	GAN
Neural network?	No	No	Yes
Your purpose	Conventional baseline	Conventional baseline	Main proposed method

11. How all three fit into YOUR project

This is your complete experimental setup:





This is a very good diagram to use when explaining your project.

12. How you actually implement the three methods

Your common starting point is:

$N = 784$

$M = 100$

Generate measurement matrix:

```
A = np.random.randn(M, N)
```

```
A = A / np.linalg.norm(A, axis=1, keepdims=True)
```

For an image:

```
x = image.flatten()
```

Generate measurement:

```
y = A @ x
```

Now the **same y** can be given to all three methods.

LASSO

$y \rightarrow \text{LASSO} \rightarrow \text{reconstructed image}$

SBL

$y \rightarrow \text{SBL} \rightarrow \text{reconstructed image}$

GAN

$y \rightarrow$ Generator $\rightarrow$ reconstructed image

This is important because it makes your comparison more meaningful: **all three methods start from the same compressed information.**

13. What your program is doing conceptually

Your entire program can be understood with just this:

Original

$$x \in R^{784}$$

Compression

$$y = Ax$$

where:

$$A \in R^{100 \times 784}$$

Therefore:

$$y \in R^{100}$$

You have:

784 numbers

↓

100 numbers

Now you need to recover:

100 numbers

↓

784 numbers

That is the **reconstruction problem.**

14. The difference in how they recover it

LASSO asks:

"Which sparse coefficients can explain my measurements?"

$$\min_s \frac{1}{2} \|y - Bs\|_2^2 + \lambda \|s\|_1$$

SBL asks:

"Which coefficients are statistically important, and what values are most probable?"

It estimates the coefficients and their relevance parameters repeatedly.

GAN asks:

"Can I learn a mapping from compressed measurements to the original image?"

$$G(y) \rightarrow \hat{x}$$

The generator learns this mapping during training.

15. One very simple way to explain it to your guide

You can say:

"I first generate compressed measurements from the original MNIST image using $y = Ax$. Then I use three different reconstruction approaches. LASSO uses L1-regularized optimization to obtain a sparse representation. SBL uses a Bayesian approach and iteratively estimates the important sparse coefficients. My GAN uses a neural network to directly learn the mapping from the compressed measurements to the reconstructed image. Finally, I compare their reconstruction errors using MSE and NMSE."

That is a strong explanation of exactly what you are implementing

Yes. For your project, think of **LASSO and SBL as two conventional reconstruction methods** that take the same compressed measurements that your GAN takes.

Your overall idea is:

Original image $\rightarrow$ measurement matrix $\rightarrow$ compressed measurements $\rightarrow$ reconstruction

For MNIST:

784 pixels $\rightarrow M$ measurements $\rightarrow$ 784 reconstructed pixels

For your current experiments, $M = 100$ or $M = 50$.

1. What is LASSO?

LASSO = Least Absolute Shrinkage and Selection Operator.

In your project, LASSO is used to recover the original image from fewer measurements by assuming that the image has a **sparse representation**.

Simple idea

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where:

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The problem is:

We know A and y , but we don't know x . How can we recover x ?

LASSO solves this optimization problem.

$$\hat{s} = \arg \min_s \left[\frac{1}{2} \|y - Bs\|_2^2 + \lambda \|s\|_1 \right]$$

where, in your DCT-based implementation,

$$B = AD$$

and

$$x = Ds$$

Here is the sparse representation of the image.

2. Why do we use DCT with LASSO?

An image itself may not be very sparse.

But after converting the image into a suitable transform domain such as **DCT**, many coefficients can become small or close to zero.

So:

$$x \rightarrow DCT \rightarrow s$$

and we try to recover s .

Then:

$$s \rightarrow IDCT \rightarrow \hat{x}$$

So your LASSO pipeline is:

Original MNIST image

↓

DCT

↓

Sparse coefficients s

↓

Measurement matrix A

↓

Measurements y

↓

LASSO

↓

Recovered coefficients

↓

IDCT

↓

Reconstructed image

3. What does the L1 term do?

This part is important:

$$\lambda \|s\|_1$$

The L1 term encourages many coefficients to become **zero or very small**.

For example:

Before LASSO:

0.8 0.02 0.7 0.001 0.03 0.6 0.004 ...

After sparse recovery:

0.8 0 0.7 0 0 0.6 0 ...

That is why LASSO is useful for **sparse recovery**.

In one sentence for your presentation:

LASSO reconstructs the image by finding a sparse representation whose measurements match the observed measurements while using L1 regularization to promote sparsity.

4. How do you implement LASSO in your program?

Your program basically does these steps:

Step 1 — Read MNIST

x = image

Shape:

28 × 28

Flatten:

784

Step 2 — Apply DCT

```
s = dct(x)
```

Now you have the sparse representation.

Step 3 — Generate measurements

```
y = A @ x
```

But because you're using DCT:

$$y = A(Ds)$$

so:

$$y = (AD)s$$

Define:

```
B = A @ D
```

Then:

```
y = B @ s
```

Step 4 — LASSO solves for s

In Python, you can use:

```
from sklearn.linear_model import Lasso
```

```
model = Lasso(alpha=0.001, max_iter=1000)
```

```
model.fit(B, y)
```

```
s_hat = model.coef_
```

Step 5 — Convert back to image

```
x_hat = idct(s_hat)
```

Now `x_hat` is your reconstructed image.

5. What is SBL?

SBL = Sparse Bayesian Learning.

SBL also tries to recover a sparse signal from fewer measurements.

But its approach is different from LASSO.

LASSO

Uses:

L1 optimization

SBL

Uses:

Bayesian probability/statistical modeling

So:

Method Main idea

LASSO L1 optimization

SBL Bayesian sparse learning

GAN Neural-network-based reconstruction

6. Simple idea of SBL

Again we start with:

$$y = Bs$$

We don't know s .

SBL assumes that every coefficient of s has a probability distribution.

For example:

$$s_i \sim N(0, \gamma_i)$$

Here:

$$\gamma_i$$

represents how important that coefficient is.

If a coefficient is not useful:

$$\gamma_i \rightarrow 0$$

Therefore:

$$s_i \rightarrow 0$$

So SBL gradually identifies which coefficients are important.

7. How SBL works step-by-step

Suppose you have:

Measurements y

↓

Initial gamma values

↓

Calculate covariance

↓

Estimate coefficients

↓

Update gamma

↓

Repeat

↓

Final sparse coefficients

↓

IDCT

↓

Reconstructed image

The important idea is:

SBL repeatedly estimates the signal and updates the importance of each coefficient.

8. What happens inside your SBL code?

Your implementation has:

```
gamma = np.ones(n)
```

Initially, all coefficients are given equal importance.

Then:

```
C = B @ Gamma @ B.T + noise * I
```

In your simplified implementation, this is represented using:

```
BG = B * gamma[np.newaxis, :]
```

```
C = BG @ B.T + noise * np.eye(m)
```

Then the program estimates the coefficients:

```
mu = gamma * (B.T @ C_inv_y)
```

Then it calculates updated values:

```
gamma_new = mu**2 + Sigma
```

Then:

```
gamma = gamma_new
```

This is repeated.

For example:

Iteration 1

↓

estimate coefficients

↓

update gamma

Iteration 2

↓

estimate coefficients

↓

update gamma

Iteration 3

↓

estimate coefficients

↓

update gamma

...

Eventually, the algorithm produces:

$$\hat{s}$$

Then:

$$\hat{x} = D\hat{s}$$

or using IDCT:

`x_hat = idct(s_hat)`

9. Very important: Is SBL adding noise?

In your current code you have:

`noise = 0.001`

This **does not mean you are randomly adding noise to your MNIST image or measurement vector.**

You use it here:

`C = BG @ B.T + noise * np.eye(m)`

It provides a small diagonal term in the covariance calculation.

So when your guide asks:

"Are you adding noise?"

You can say:

"No. In my current implementation, I am not adding random noise to the measurement data. The 0.001 parameter is used as a small noise or regularization term inside the SBL covariance calculation."

10. LASSO vs SBL vs GAN

This is the most important comparison for your project.

Feature	LASSO	SBL	GAN
Type	Optimization	Bayesian	Deep learning
Main principle	L1 sparsity	Bayesian sparsity	Generator + discriminator
Input	Measurements	Measurements	Measurements
Output	Reconstructed image	Reconstructed image	Reconstructed image
Iterative?	Yes, internally	Yes	Training epochs
Uses L1?	Yes	No	No
Uses probability model?	No	Yes	Not in the same sparse Bayesian sense
Neural network?	No	No	Yes
Your purpose	Conventional baseline	Conventional baseline	Main proposed method

11. How all three fit into YOUR project

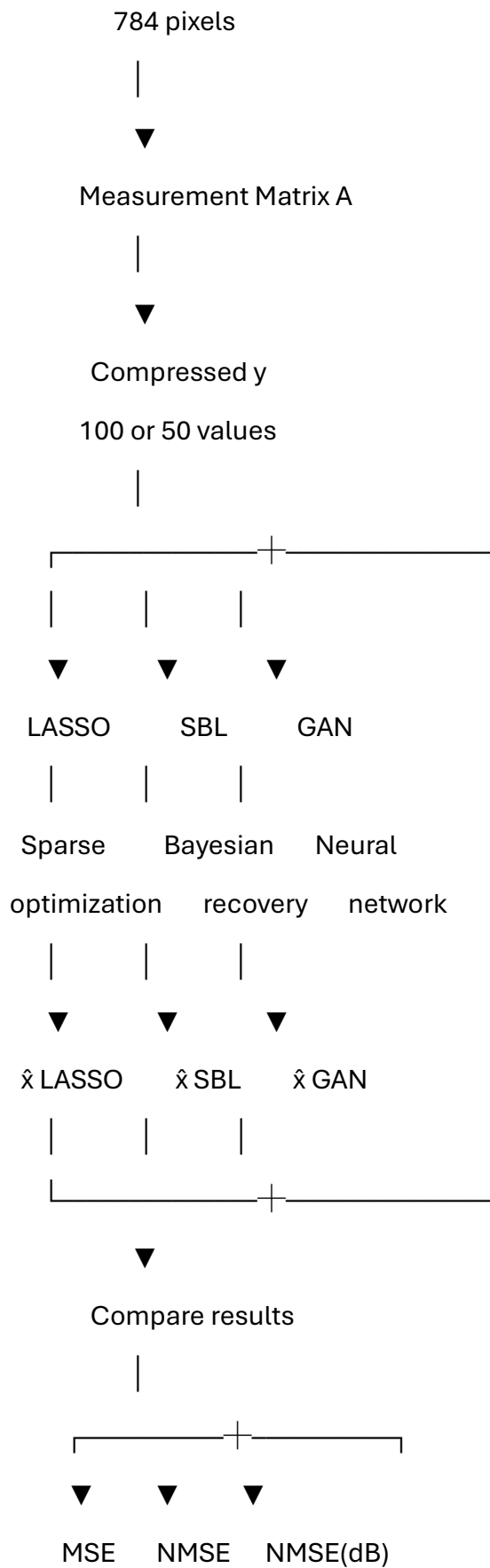
This is your complete experimental setup:

ORIGINAL IMAGE

|

MNIST 28×28

|



This is a very good diagram to use when explaining your project.

12. How you actually implement the three methods

Your common starting point is:

$N = 784$

$M = 100$

Generate measurement matrix:

$A = \text{np.random.randn}(M, N)$

$A = A / \text{np.linalg.norm}(A, \text{axis}=1, \text{keepdims}=\text{True})$

For an image:

$x = \text{image.flatten}()$

Generate measurement:

$y = A @ x$

Now the **same** y can be given to all three methods.

LASSO

$y \rightarrow \text{LASSO} \rightarrow \text{reconstructed image}$

SBL

$y \rightarrow \text{SBL} \rightarrow \text{reconstructed image}$

GAN

$y \rightarrow \text{Generator} \rightarrow \text{reconstructed image}$

This is important because it makes your comparison more meaningful: **all three methods start from the same compressed information.**

13. What your program is doing conceptually

Your entire program can be understood with just this:

Original

$$x \in R^{784}$$

Compression

$$y = Ax$$

where:

$$A \in R^{100 \times 784}$$

Therefore:

$$y \in R^{100}$$

You have:

784 numbers

↓

100 numbers

Now you need to recover:

100 numbers

↓

784 numbers

That is the **reconstruction problem**.

14. The difference in how they recover it

LASSO asks:

"Which sparse coefficients can explain my measurements?"

$$\min_s \frac{1}{2} \|y - Bs\|_2^2 + \lambda \|s\|_1$$

SBL asks:

"Which coefficients are statistically important, and what values are most probable?"

It estimates the coefficients and their relevance parameters repeatedly.

GAN asks:

"Can I learn a mapping from compressed measurements to the original image?"

$$G(y) \rightarrow \hat{x}$$

The generator learns this mapping during training.

15. One very simple way to explain it to your guide

You can say:

"I first generate compressed measurements from the original MNIST image using $y = Ax$. Then I use three different reconstruction approaches. LASSO uses L1-regularized optimization to obtain a sparse representation. SBL uses a Bayesian approach and iteratively estimates the important sparse coefficients. My GAN uses a neural network to directly learn the mapping from the compressed measurements to the reconstructed image. Finally, I compare their reconstruction errors using MSE and NMSE."

That is a strong explanation of exactly what you are implementing

1. What is sparse recovery?

Answer:

Sparse recovery means **recovering the original signal from a small number of measurements by assuming that the signal has a sparse representation.**

In my project, the original MNIST image has:

$$28 \times 28 = 784$$

pixels. I obtain fewer measurements, such as **100 or 50**, and try to reconstruct the original 784-pixel image.

The basic model is:

$$y = Ax$$

where x is the original image, A is the measurement matrix, and y is the compressed measurement.

Short answer:

Sparse recovery is the process of recovering a signal from fewer measurements by exploiting sparsity in the signal or its representation.

2. What is compressed sensing?

Answer:

Compressed sensing is a technique for acquiring a signal using **fewer measurements than the original number of signal values** and then reconstructing the signal from those measurements.

In my project:

784 pixels $\rightarrow$ 100 measurements

or

784 pixels $\rightarrow$ 50 measurements

The measurements are generated using:

$$y = Ax$$

Then LASSO, SBL, or GAN is used to reconstruct the original image.

Short answer:

Compressed sensing allows us to acquire fewer measurements of a signal and reconstruct the original signal by using prior information such as sparsity.

3. What does $y = Ax$ mean?

This is **very important** for your project.

$$y = Ax$$

means that the original signal x is transformed into a smaller set of measurements y using the measurement matrix A .

For your MNIST image:

$$x = 784 \times 1$$

If you use 100 measurements:

$$A = 100 \times 784$$

Therefore:

$$y = Ax$$

gives:

$$y = 100 \times 1$$

So:

Original image x

784 values

↓

Measurement matrix A

100×784

↓

Measurements y

100 values

Short answer:

$y = Ax$ represents the measurement process. x is the original image, A is the measurement matrix, and y contains the compressed measurements.

4. Why can 100 measurements reconstruct a 784-pixel image?

Answer:

Normally, 100 measurements are not enough to uniquely determine an arbitrary 784-dimensional signal.

However, image reconstruction methods use **additional assumptions or learned information**.

In my project, the methods exploit different forms of prior information:

- **LASSO** → sparsity using L1 regularization
- **SBL** → Bayesian sparse representation
- **GAN** → learns the relationship between measurements and images from training data

Therefore, the goal is not to recover an arbitrary 784-value vector, but to reconstruct an image using its structure or learned characteristics.

Good viva answer:

Although 100 measurements are fewer than 784 pixels, reconstruction is possible because the methods use prior information about the signal, such as sparsity or learned image structure.

5. What is LASSO and why does it use L1?

Answer:

LASSO stands for:

Least Absolute Shrinkage and Selection Operator.

It is an optimization-based sparse recovery method.

The objective is:

$$\hat{s} = \arg \min_s \left[\frac{1}{2} \| y - Bs \|_2^2 + \lambda \| s \|_1 \right]$$

There are two important parts.

First part:

$$\frac{1}{2} \| y - Bs \|_2^2$$

This makes the reconstructed measurements close to the actual measurements.

Second part:

$$\lambda \| s \|_1$$

This encourages the solution to be sparse.

So LASSO tries to find a solution that:

matches the measurements while keeping the representation sparse.

In my implementation, I use the DCT representation and `sklearn.linear_model.Lasso`.

Short viva answer:

LASSO uses L1 regularization to encourage sparsity while minimizing the difference between the measured signal and the reconstructed signal.

6. What is SBL and how is it different from LASSO?

Answer:

SBL stands for:

Sparse Bayesian Learning.

It is a Bayesian approach for sparse signal recovery.

Instead of directly using L1 regularization like LASSO, SBL assigns statistical parameters to the coefficients and iteratively estimates which coefficients are important.

The basic process is:

Initialize parameters

↓

Calculate covariance

↓

Estimate coefficients

↓

Update parameters

↓

Repeat

↓

Final sparse representation

Difference:

LASSO	SBL
Optimization-based	Bayesian-based
Uses L1 regularization	Uses statistical/Bayesian modeling
Controls sparsity using λ	Controls coefficient relevance using parameters such as γ

LASSO

SBL

Uses an optimization solver Iteratively estimates coefficients and parameters

Short viva answer:

LASSO obtains sparse solutions using L1 regularization, whereas SBL obtains sparse solutions using a Bayesian iterative estimation process.

7. What is a GAN and how is your GAN different from a normal GAN?

Answer:

GAN stands for:

Generative Adversarial Network.

A GAN has two main networks:

1. **Generator**
2. **Discriminator**

The Generator produces an image, while the Discriminator tries to distinguish between real and generated images.

In a conventional GAN, the generator commonly receives random noise z :

$$G(z) \rightarrow \text{image}$$

But in **my project**, the Generator receives the compressed measurement y :

$$G(y) \rightarrow \hat{x}$$

So my GAN is being used for **measurement-conditioned image reconstruction**.

The basic idea is:

Compressed measurements

↓

Generator

↓

Reconstructed image

↓

Discriminator

↓

Real or generated?

I also use reconstruction loss so that the reconstructed image remains close to the original image.

Short viva answer:

My GAN is different from a conventional GAN because the Generator receives compressed measurements instead of only random noise, and its purpose is image reconstruction rather than simply generating random images.

8. Why does reconstruction error increase when measurements decrease from 100 to 50?

Answer:

When the number of measurements decreases:

$100 \rightarrow 50$

the amount of information available about the original image also decreases.

Therefore the reconstruction problem becomes more difficult.

Conceptually:

784 $\rightarrow$ 100 measurements

↓

More information

↓

Better reconstruction

while:

784 $\rightarrow$ 50 measurements

↓

Less information

↓

More difficult reconstruction

So we generally expect the reconstruction error to increase.

This is consistent with your experiments where the **784→50 case has higher reconstruction error than the 784→100 case.**

Short viva answer:

Reducing the measurements provides less information about the original image, making reconstruction more difficult and generally increasing the reconstruction error.

9. What are MSE, NMSE and NMSE(dB)?

MSE

Mean Squared Error measures the average squared difference between the original and reconstructed pixels.

$$MSE = \frac{1}{N} \sum_{i=1}^N (x_i - \hat{x}_i)^2$$

Lower MSE means the reconstruction is closer to the original.

NMSE

Normalized Mean Squared Error normalizes the reconstruction error relative to the original signal energy.

$$NMSE = \frac{\|x - \hat{x}\|_2^2}{\|x\|_2^2}$$

Again, lower is better.

NMSE(dB)

For easier comparison, NMSE can be represented in decibels:

$$NMSE_{dB} = 10 \log_{10}(NMSE)$$

For example:

$$NMSE = 0.3$$

gives approximately:

$$NMSE_{dB} = -5.23 \text{ dB}$$

So:

Lower NMSE is better, and a more negative NMSE(dB) indicates lower normalized error.

10. How exactly does your Python program go from MNIST → measurements → LASSO/SBL/GAN → reconstructed image?

This is the **most important answer** because it explains your entire implementation.

Step 1 — Load MNIST

The MNIST image is:

$$28 \times 28$$

Therefore:

$$28 \times 28 = 784$$

pixels.

Step 2 — Normalize

Pixel values are converted to approximately:

$$0 \leq x_i \leq 1$$

Step 3 — Flatten

The image is converted:

$$28 \times 28$$

↓

$$784 \times 1$$

So:

$$x \in R^{784}$$

Step 4 — Create measurement matrix

For 100 measurements:

$$A \in R^{100 \times 784}$$

Step 5 — Generate measurements

The program calculates:

$$y = Ax$$

Therefore:

784 values

↓

measurement matrix

↓

100 values

Step 6 — LASSO reconstruction

LASSO uses the measurements to estimate the sparse representation:

$$\hat{s}_{LASSO}$$

Then the representation is converted back to the image:

$$\hat{x}_{LASSO}$$

Step 7 — SBL reconstruction

SBL takes the same measurement information and iteratively estimates the sparse coefficients:

$$\hat{s}_{SBL}$$

Then reconstructs:

$$\hat{x}_{SBL}$$

Step 8 — GAN reconstruction

The same measurement y is given to the trained Generator:

$$\hat{x}_{GAN} = G(y)$$

The output has 784 values, which are reshaped into:

$$28 \times 28$$

Step 9 — Compare

Finally, compare:

Original image

|

└─ LASSO reconstruction

|

└─ SBL reconstruction

|

└─ GAN reconstruction

using:

- MSE
 - NMSE
 - NMSE(dB)
-

Your complete answer in one flow

If your guide says:

"Explain your project technically."

You can answer:

"My project focuses on reconstructing an image from a reduced number of measurements. I use MNIST images of size 28×28 , so each image contains 784 pixels. I generate a smaller measurement vector using the measurement model $y = Ax$, where A is the measurement matrix. I then reconstruct the image using three approaches: LASSO, SBL, and a measurement-conditioned GAN. LASSO uses L1-regularized optimization to obtain a sparse representation. SBL uses Bayesian iterative estimation to identify important coefficients. The GAN learns a mapping from the compressed measurements to the original image. Finally, I compare the reconstructed images using MSE, NMSE, and NMSE in dB. I also compare different measurement levels, such as 100 and 50 measurements, to study how measurement reduction affects reconstruction quality."

That is the **core explanation of your entire project**.

Yes. I reviewed the **actual PPT you uploaded**, not just the earlier outline. Your presentation already has the main project story, but there are a few things I would fix before you present it.

First: what is missing / what should be improved?

Your current PPT has these sections:

- Introduction
- Problem Statement
- Literature/reference papers
- Initial DNN learning
- GAN learning
- Software/libraries
- Dataset
- Conventional comparison
- Future work
- Experimental results
- Performance evaluation
- References

That is a reasonable structure. But I recommend **three important changes** before presenting:

1. Add a clear "Objective" slide

You currently go from Introduction → Problem Statement. Add:

Project Objectives

- To study image reconstruction from compressed measurements.
- To understand the measurement model $y = Ax$.
- To develop a GAN-based image reconstruction model.
- To study conventional reconstruction methods such as LASSO and SBL for comparison.
- To evaluate reconstruction quality using MSE and NMSE.
- To test the reconstruction approach on additional datasets in future work.

This makes it very clear to your guide **what exactly you are trying to achieve**.

2. Your current Problem Statement says $M = 100$

Your PPT says:

Image dimension = 784

Measurement dimension = 100

and therefore:

$784 \rightarrow 100$.

Later your results also show **$784 \rightarrow 100$** and **$784 \rightarrow 50$** .

So when presenting, **do not suddenly say $M = 196$** unless you change the PPT and results. For this PPT, stay consistent with:

$784 \rightarrow 100$

and experimental comparison:

$784 \rightarrow 100$ and $784 \rightarrow 50$

That is particularly important because your guide may immediately notice an inconsistency.

3. Your present-status section needs to be explained better

Your PPT has a "Present - Conventional Comparison" section and then LASSO/SBL. It correctly says they are comparison baselines rather than the main project.

You should verbally make this very clear:

"My main focus is GAN-based reconstruction. LASSO and SBL are implemented as conventional comparison methods so that I can evaluate the GAN reconstruction against established sparse-recovery approaches."

That one sentence will prevent a lot of confusion.

Now, how should you PRESENT the whole PPT?

Don't read the bullet points. **Explain the story.**

Below is what I recommend you actually say.

Slide 1 — Title

What you say

"Good morning/afternoon everyone.

I am Aravind Ashok, M.Tech Computer Science and Engineering student.

Today I am presenting the current progress of my project titled **Sparse-Measurement Image Reconstruction**.

The main objective of my project is to reconstruct an image from a smaller number of measurements using a GAN-based approach, and to compare the reconstruction with conventional methods."

Then pause.

Don't start explaining equations here.

Slide 2 — Contents

Your PPT currently lists the major sections.

Say:

"In this presentation, I will first explain the problem and the reference paper I studied. Then I will explain my initial learning of DNN and GAN, the software and dataset used, the reconstruction approach, the present comparison with conventional methods, the experimental results, and finally my future work."

This gives the audience a roadmap.

Slide 3 — Introduction

Your slide explains image reconstruction and the basic idea of using fewer measurements.

Say:

"Image reconstruction means recovering an image when we do not have all the original information available.

In my project, instead of directly providing all 784 pixels of an MNIST image to the reconstruction system, I first convert the image into a smaller number of measurements.

The reconstruction model then uses these measurements to recover the original image.

My main approach is GAN-based reconstruction, and I use LASSO and SBL as conventional comparison methods."

If they ask:

"Why do you need reconstruction?"

Say:

"The purpose is to study whether useful image information can be recovered even when only a smaller amount of measurement information is available."

Slide 4 — Problem Statement

Your current PPT says:

$28 \times 28 = 784$ pixels

and **100 measurements**.

Say:

"An MNIST image contains 784 pixel values. If I use all 784 values, there is no measurement reduction.

So in my experiment, I reduce the representation to 100 measurements.

Therefore, the reconstruction problem becomes 100 measurements to 784 pixels.

Mathematically, I represent this as y equals A times x ."

Then explain:

"Here x represents the original image, A is the measurement matrix, and y represents the compressed measurements."

Very likely question:

Q: Is this actually image compression?

Answer carefully:

"It is measurement-based dimensionality reduction for reconstruction. I am not implementing a conventional JPEG compression algorithm. I generate fewer measurements and then reconstruct the original image from them."

That's a strong answer.

ADD THIS — Slide 5: Project Objectives

I strongly recommend adding this.

What you say:

"Based on this problem, I defined the following objectives.

First, to understand sparse measurement and image reconstruction.

Second, to study DNN and GAN-based reconstruction.

Third, to implement a GAN-based reconstruction model.

Fourth, to compare the reconstruction with conventional sparse-recovery methods such as LASSO and SBL.

Finally, I plan to test the models with additional datasets."

Slide 6 — Literature / Reference Paper

Your PPT identifies **Sparse Signal Recovery Using Deep Neural Networks** as the main paper.

Say:

"The main reference paper I studied is Sparse Signal Recovery Using Deep Neural Networks by Rubin Jose Peter and Chandra R. Murthy.

The paper studies recovery of signals from sparse measurements.

This paper helped me understand the basic problem formulation of recovering a signal from fewer measurements.

Based on this concept, I am studying a GAN-based approach for image reconstruction."

Important:

Don't say:

"I reproduced the exact paper."

Instead say:

"The paper is my main reference and provides the conceptual foundation for my project."

That's safer and more accurate.

Slide 7 — Literature Review

Your PPT includes additional GAN/compressed-sensing papers, including 2024 and 2025 works.

Say:

"Apart from the main reference paper, I also studied recent work related to GAN-based image reconstruction and compressed sensing.

These papers helped me understand how GANs are being applied to reconstruction problems in areas such as single-pixel imaging, MRI reconstruction and CT reconstruction.

My current implementation is focused on MNIST as an initial experimental dataset."

Question:

"Why are you using MNIST instead of MRI or CT?"

Answer:

"MNIST is a simple dataset that allows me to first verify the reconstruction methodology and compare the models under controlled conditions. After validating the approach, my next step is to test additional datasets."

This connects directly to your future work.

Slide 8 — Overall Reconstruction Pipeline

This is one of the **most important slides**.

Your PPT has an overall reconstruction pipeline slide.

Say slowly:

"This is the overall workflow of my project.

First, I take the original MNIST image.

The image is represented as a 784-dimensional vector.

Then I apply the measurement matrix to obtain a smaller measurement vector.

In my current experiment, 784 values are reduced to 100 measurements.

These measurements are then given to the reconstruction model.

The model reconstructs a 784-dimensional image.

Finally, I compare the reconstructed image with the original image using reconstruction error metrics."

Then point to each block while speaking.

Slide 9 — Initial Learning: DNN

Your PPT says you first studied DNN fundamentals such as layers, neurons, activation functions, loss and training.

Say:

"Before implementing the GAN, I first studied the basic concepts of Deep Neural Networks.

I learned about the input, hidden and output layers, neurons, weights, activation functions, loss functions and model training.

This gave me the basic understanding needed to move toward GAN-based reconstruction."

Expected question:

"Why did you study DNN if your final model is GAN?"

Answer:

"GANs are built using neural networks, so understanding the basic DNN structure helped me understand how the Generator and Discriminator are constructed and trained."

Slide 10 — Learning About GAN

This slide should be important because GAN is your main model.

Say:

"After learning the basic DNN concepts, I studied Generative Adversarial Networks.

A GAN contains two main networks: the Generator and the Discriminator.

The Generator produces the reconstructed image, while the Discriminator tries to distinguish between real and generated images.

In my project, instead of giving random noise as the main input, I use the compressed measurement as the input to the reconstruction Generator."

Very likely question:

"Is your GAN a normal GAN?"

Good answer:

"It is a reconstruction-oriented or measurement-conditioned GAN. The Generator is conditioned on the compressed measurement rather than using only random noise."

Slide 11 — Software, Libraries and Functions

Your PPT includes this section.

Say:

"For implementation, I used Python as the main programming language.

NumPy was used for numerical and matrix operations.

TensorFlow and Keras were used to construct and train the GAN.

SciPy was used for mathematical operations such as DCT where required.

scikit-learn was used for conventional reconstruction methods such as LASSO.

Matplotlib was used for plotting and visualization."

If asked about functions, mention only ones you actually used:

- numpy
- matrix multiplication
- reshape
- TensorFlow/Keras Dense
- Input

- Model
 - Adam
 - loss functions
 - Matplotlib plotting
 - LASSO from scikit-learn
-

Slide 12 — Dataset and Experimental Setup

Your PPT uses MNIST and says 28×28 , 784 values, normalized 0–1.

Say:

"For the initial experiment, I used the MNIST handwritten digit dataset.

Each image has a size of 28 by 28, giving 784 pixel values.

The pixel values are normalized between 0 and 1.

I then generate a lower-dimensional measurement representation and use it for reconstruction."

Question:

"Why normalize the image?"

Answer:

"Normalization brings the pixel values into a consistent numerical range, which makes neural-network training more stable."

Slide 13 — Present Status: Conventional Comparison

This is where you need to be very clear.

Your PPT states that LASSO and SBL are conventional baselines.

Say:

"At the present stage, I have moved to comparison with conventional reconstruction methods.

I selected LASSO and Sparse Bayesian Learning as comparison methods.

LASSO is an optimization-based sparse-recovery method using L1 regularization.

SBL is a Bayesian sparse-recovery method.

I am not treating these as the main contribution of my project. They are used as reference methods to compare the GAN reconstruction."

Very likely question:

"Why did you choose LASSO and SBL?"

Answer:

"Because they represent two different conventional approaches to sparse recovery: LASSO uses optimization with L1 regularization, while SBL uses a Bayesian approach. This gives me useful conventional baselines for evaluating the GAN."

Slide 14 — Experimental Results: 784 → 100

Your PPT reports:

N = 784 → 100

Normalized MSE = 0.12024844

Say:

"Here I tested the reconstruction with 100 measurements from an original 784-dimensional image.

The normalized MSE obtained in this experiment is approximately 0.1202.

The reconstructed image shows that the model is able to recover the basic structure of the handwritten digit from significantly fewer measurements."

Don't say "excellent" or "very good" unless your guide asks you to interpret it.

Slide 15 — Experimental Results: 784 → 50

Your PPT reports:

784 → 50

Normalized MSE = 0.30164862

Say:

"I also reduced the number of measurements further, from 100 to 50.

In this case, the normalized MSE increased to approximately 0.3016.

This indicates that reconstruction becomes more difficult when fewer measurements are available."

This is an excellent observation to tell your guide.

It shows you understand your experiment.

Slide 16 — Performance Evaluation

Your PPT has MSE, NMSE and NMSE(dB).

Say:

"To quantitatively evaluate reconstruction quality, I use three measures.

The first is MSE, which measures the average squared difference between the original and reconstructed pixels.

The second is NMSE, which normalizes the reconstruction error with respect to the original signal energy.

Finally, NMSE can be expressed in decibels using $10 \log_{10}$ of NMSE.

These metrics allow me to compare reconstruction quality numerically."

Expected question:

"Why use NMSE instead of only MSE?"

Answer:

"MSE depends on the scale and energy of the signal. NMSE normalizes the error relative to the original signal energy, which makes it more useful when comparing reconstruction results."

Slide 17 — Future Work

Your PPT already has a good future-work section.

Say:

"My next work is to move beyond MNIST and test the reconstruction framework using additional datasets.

I plan to preprocess the new datasets, generate compressed measurements, and test the GAN, LASSO and SBL methods.

Then I will compare their reconstruction performance under different measurement conditions.

I also plan to study how the number of measurements affects reconstruction quality."

Slide 18 — References

Your references include the main paper and additional reconstruction/GAN papers.

Say:

"These are the main references I used for understanding sparse signal recovery, compressed sensing and GAN-based image reconstruction."

Don't spend more than 20–30 seconds here.

Slide 19 — Thank You

Say:

"That concludes my current project progress. Thank you. I am happy to answer any questions."

Then **stop talking**.

Don't immediately start explaining something else.

Questions your GUIDE is very likely to ask

Prepare these especially well.

1. What exactly is your project?

Answer:

"My project studies image reconstruction from a smaller number of measurements. I use a measurement model to reduce a 784-dimensional MNIST image to a lower-dimensional measurement and then reconstruct the original image using a GAN. I compare the reconstruction with conventional methods such as LASSO and SBL."

2. What is the novelty of your project?

Be careful here.

Don't claim that GAN reconstruction itself is completely new.

Say:

"The current work focuses on applying a measurement-conditioned GAN reconstruction framework to the sparse-measurement image reconstruction problem and

experimentally comparing it with conventional sparse-recovery methods. The present implementation is an experimental study, and I plan to extend it to additional datasets."

3. What is sparse recovery?

"Sparse recovery is the process of recovering an unknown signal from a smaller number of measurements by using prior information such as sparsity."

4. What is $y = Ax$?

" x is the original signal or image, A is the measurement matrix, and y is the smaller set of measurements obtained from x ."

5. Why is this problem difficult?

"Because the number of measurements is smaller than the number of unknown image values. Therefore, the system is underdetermined and there can be many possible signals that produce similar measurements."

6. Why GAN?

"I studied GANs because they can learn a mapping from the available measurements to the desired image reconstruction, while adversarial training can encourage realistic image outputs."

7. Why not use only DNN?

"DNN was part of my initial learning stage. I then studied GANs and selected a GAN-based reconstruction approach as the main model for my current project."

8. What is the Generator doing?

"The Generator takes the compressed measurement as input and produces the reconstructed 784-pixel image."

9. What is the Discriminator doing?

"The Discriminator tries to distinguish between real images and reconstructed images, providing adversarial feedback to the Generator during training."

10. Why LASSO?

"LASSO provides a conventional sparse-recovery baseline and uses L1 regularization to encourage sparse solutions."

11. Why SBL?

"SBL provides a Bayesian sparse-recovery baseline, so I can compare the learning-based GAN with a different conventional sparse-recovery approach."

12. Why compare with conventional methods?

"Without comparison, I can only show that my GAN works. By comparing with conventional methods under the same measurement conditions, I can evaluate its reconstruction behaviour relative to established approaches."

13. Why MNIST?

"MNIST is a relatively simple and well-defined dataset, so it is suitable for initially validating the reconstruction methodology before moving to more complex datasets."

14. Why 100 measurements?

"I wanted to test reconstruction using substantially fewer values than the original 784 pixels. I selected 100 measurements for the current experiment and also tested 50 measurements to study the effect of further reducing the measurement dimension."

15. What happens when you reduce 100 to 50?

This one you already have experimental evidence for.

"The reconstruction becomes more difficult. In my results, the normalized MSE increased from approximately 0.1202 for 100 measurements to approximately 0.3016 for 50 measurements."

This is one of your **strongest answers** because it comes directly from your experiment.

16. What does lower MSE mean?

"Lower MSE means the reconstructed pixels are, on average, closer to the original pixels."

17. What does NMSE mean?

"NMSE is the reconstruction error normalized by the energy of the original signal."

18. Why dB?

"The logarithmic dB representation makes relative error values easier to compare, particularly when the values span different scales."

19. Is your current result good?

Don't simply say "yes."

Say:

"The current results show that reconstruction is possible from reduced measurements. However, I consider this an initial experimental result. The next step is to test the approach on additional datasets and analyze the performance more comprehensively."

That sounds much more research-oriented.

Questions they may ask about your future work**"Which dataset will you use next?"**

If you haven't finalized it, **don't invent one.**

Say:

"I have not finalized the next dataset yet. I plan to select an image dataset with greater complexity than MNIST and evaluate the same reconstruction framework."

"Why do you need another dataset?"

"MNIST is relatively simple. Testing additional datasets will help determine whether the reconstruction approach generalizes beyond simple handwritten digits."

"What will you compare?"

"I plan to compare GAN, LASSO and SBL using the same measurement setup and evaluation metrics."

One very important thing before your presentation

Your current PPT has a **potentially confusing transition**:

Introduction → DNN → GAN → Software → Dataset → LASSO/SBL → Results

I recommend adding one slide immediately after the literature review:

Project Objectives

Then your story becomes:

Problem

↓

Reference Paper

↓

What I learned initially: DNN

↓

What I learned next: GAN

↓

My project methodology

↓

Implementation

↓

Conventional comparison

↓

Results

↓

Future work

That will make your presentation much easier to follow.

Also remember these 5 numbers/concepts

Before you go into the presentation, be completely comfortable with:

$$28 \times 28 = 784$$

$$784 \rightarrow 100$$

$$784 \rightarrow 50$$

$$y = Ax$$

$$NMSE = \frac{\|x - \hat{x}\|^2}{\|x\|^2}$$

If your guide asks you something unexpected, come back to these concepts. They are the core of your current experiment.

And most importantly, **do not memorize the entire script word-for-word**. Understand the flow and use the slide as a prompt. Your presentation will sound much more natural that way.

Yes. Since this is a **project review presentation**, you should not simply read the bullet points. You should explain the slides as a story: **what you learned, why you learned it, and how you used it in your project**.

Here is a simple **slide-by-slide speaking script** based exactly on your 15 slides.

Slide 1 — Title

You can start:

“Good morning/afternoon everyone. I am Aravind Ashok from the Department of Computer Science and Engineering. Today I am presenting the progress of my M.Tech project, titled ‘High-Resolution Image Reconstruction Using GAN.’ In this presentation, I will explain the concepts I have studied, the methods I have implemented, the experiments I have conducted, and the current status and future work of my project.”

Then move to the introduction.

Slide 2 — Introduction / Background

Say:

“First, I started by understanding the basic problem of image reconstruction. An image contains a large amount of information because every pixel represents a value. Instead of using the complete image, we can take a smaller number of measurements and try to reconstruct the original image from those measurements.”

Then point to the equation:

“The basic measurement model I studied is $y = Ax$. Here, x represents the original image, A is the measurement matrix, and y represents the measurements obtained from the original image.”

Then:

“The main idea of my project is to take these low-dimensional measurements and reconstruct the original image using a GAN-based approach.”

Slide 3 — Problem Statement

Say:

“After understanding the basic concept, I identified the main problem. If the original image has 784 values but we use only 196 measurements, we have less information than the original signal. Therefore, directly recovering the original image becomes an underdetermined problem.”

Point to:

$$M < N$$

Then:

“So the objective of my project is to investigate whether a GAN can learn the relationship between these reduced measurements and the original image, and then reconstruct the image from the measurements.”

Then mention comparison:

“I also wanted to compare this learning-based approach with conventional sparse recovery methods, specifically LASSO and SBL.”

Slide 4 — Reference Paper

This is where you explain your connection to the paper.

Say:

“To understand this problem, I first studied the reference paper ‘Sparse Signal Recovery Using Deep Neural Networks’ by Rubin Jose Peter and Chandra R. Murthy from IISc.”

Then:

“From this paper, I learned the basic concept of sparse signal recovery, the measurement model, and how deep-learning-based methods can be used for recovering signals from limited measurements.”

Then make the distinction:

“The paper provided the foundation for my study. Based on the concepts I learned, I explored a GAN-based approach for image reconstruction in my project.”

This sentence is useful because it makes clear that **you are inspired by the paper but your implementation is your project work.**

Slide 5 — Learning: Sparse Signal Recovery

Start with:

“Before implementing the model, I first had to understand what sparse signal recovery actually means.”

Then:

“In simple terms, the objective is to recover the original signal when we have only a limited number of measurements.”

Point to $y = Ax$:

“Here, the original image is represented by x . By multiplying x with the measurement matrix A , I obtain the measurement vector y . The recovery algorithm then tries to estimate x from y .”

Then explain your values:

“For my MNIST experiment, the original image is 28 by 28 pixels, so after flattening it becomes a 784-dimensional vector. I used 196 measurements, so M is less than N .”

Finally:

“This helped me understand why the reconstruction problem is difficult and why a suitable recovery method is required.”

Slide 6 — Learning: DNN and GAN

Start:

“After understanding sparse recovery, I studied how deep learning could be used for this reconstruction problem.”

Then:

“Initially, I studied the basic concepts of neural networks, such as input layers, hidden layers, weights, biases, activation functions, loss functions and optimization.”

Then move to GAN:

“After understanding the basics of neural networks, I studied GANs. A GAN mainly contains a generator and a discriminator.”

Explain simply:

“The generator produces the reconstructed image, while the discriminator tries to distinguish between real and generated images during training.”

Then connect directly to your project:

“In a conventional GAN, random noise is commonly used as input. But in my reconstruction problem, I use the measurement vector y as the input to the generator. The generator then produces the reconstructed image.”

Show:

$$y \rightarrow G(y) \rightarrow \hat{x}$$

Then:

“This was an important part of my learning because I had to understand how the GAN concept could be adapted to my image reconstruction problem.”

Slide 7 — Mathematical Formulation

Say:

“After understanding the concepts, I studied the mathematical formulation of the problem.”

Point to:

$$y = Ax$$

“This is the main measurement equation used in my project.”

Then:

“The original image x has 784 elements. A is the measurement matrix, and y contains only 196 measurements.”

Then explain reconstruction:

$$\hat{x} = G(y)$$

“The GAN tries to learn a mapping from y to the reconstructed signal $\hat{x}$.”

Then metrics:

“To measure how close the reconstructed image is to the original image, I used MSE and NMSE. I also converted NMSE into decibels to analyse the reconstruction performance.”

Don't spend too much time deriving the equations. Your guide can see them on the slide.

Slide 8 — Proposed GAN Methodology

Start:

“After studying the theory and mathematical formulation, I implemented the proposed methodology.”

Then explain the flow:

“First, I take an MNIST image of size 28 by 28. I normalize the image and flatten it into a 784-dimensional vector.”

“Next, I multiply this vector with the measurement matrix to obtain 196 measurements.”

“These measurements are then given to the GAN generator. The generator produces a 784-dimensional reconstructed image.”

“Finally, I reshape the output back to 28 by 28 and compare it with the original image.”

Then:

“The reconstruction performance is evaluated using MSE, NMSE and NMSE in dB.”

Slide 9 — GAN Architecture

Here, don't read every layer mechanically.

Say:

“This slide shows the architecture used for the GAN-based reconstruction.”

Then:

“The input to the generator is the measurement vector. In my main experiment, the input dimension is 196 and the output dimension is 784.”

“The generator uses dense layers to learn the mapping from the low-dimensional measurement space to the original image space.”

Then discriminator:

“During GAN training, the discriminator receives real and generated images and learns to distinguish between them. This adversarial process helps the generator learn to produce more realistic reconstructed images.”

Finish:

“So the main reconstruction path during inference is measurement vector → generator → reconstructed image.”

Slide 10 — Software, Libraries and Implementation

This is your **“what I actually learned by coding”** slide.

Say:

“After understanding the methodology, I implemented the experiments using Python.”

Then:

“I used NumPy for array and matrix operations, TensorFlow and Keras for developing and training the GAN model, Matplotlib for visualization, and Scikit-learn for implementing LASSO.”

Then:

“I also learned how to work with the MNIST IDX files, normalize the image data, flatten the images, perform matrix multiplication, train the model and calculate reconstruction metrics.”

Then:

“I used both VS Code and Google Colab during the implementation and testing stages.”

This slide demonstrates that you **actually worked with the implementation**, not just studied the theory.

Slide 11 — Dataset and Experimental Setup

Say:

“For the initial experiments, I used the MNIST handwritten digit dataset.”

Then:

“Each image is a 28 by 28 grayscale image, which gives 784 pixels after flattening.”

Point to parameters:

“For the main experiment, I used 10,000 training samples and 196 measurements. The batch size was 64.”

Then:

“I also experimented with reducing the number of measurements to observe how measurement reduction affects reconstruction quality.”

This is a good place to mention your $M = 100$ experiment if it is included in your results.

Slide 12 — GAN Results

Now move from **learning to actual results**.

Say:

“After implementing and training the GAN, I tested the model using MNIST images.”

Show your original/reconstructed images.

“Here, the first images represent the original input, and the reconstructed images are produced from the reduced measurements.”

Then:

“The reconstructed images retain the main structure of the handwritten digits. However, the reconstruction quality changes depending on the number of measurements.”

Then mention your observation:

“When I used 196 measurements, the reconstructed images were relatively clearer. When I reduced the number of measurements further, the reconstruction became more difficult and the error increased.”

This is a good **experimental observation** rather than simply claiming the method is better.

Slide 13 — LASSO & SBL

Say:

“After implementing the GAN, I studied conventional sparse recovery methods so that I could perform a comparative analysis.”

LASSO:

“The first method I studied was LASSO. LASSO uses L1 regularization and tries to obtain a sparse solution while minimizing the reconstruction error.”

SBL:

“The second method was Sparse Bayesian Learning, or SBL. SBL uses a Bayesian approach and performs iterative estimation of the signal parameters.”

Then:

“I implemented both methods using the same basic measurement setup and generated reconstructed images. I then evaluated them using the same types of reconstruction metrics.”

Slide 14 — Comparison

Start:

“After implementing all three approaches, I started the comparative analysis.”

Point to each:

“The three methods considered are the proposed GAN approach, LASSO and SBL.”

Then:

“The comparison is based on reconstruction quality and quantitative metrics such as MSE, NMSE and NMSE in dB.”

When showing your graph:

“This graph shows the performance of the three methods in a single plot. I am using the same evaluation framework so that the methods can be compared under the same experimental conditions.”

Then show reconstructed images:

“Along with the numerical results, I also compare the reconstructed images visually because image reconstruction quality cannot be understood completely from a single numerical value.”

Avoid saying **“GAN is the best”** unless your experimental evidence and comparison design support a carefully qualified statement. Say **“the results show differences in reconstruction performance”** and explain what your measurements show.

Slide 15 — Current Status, Learning Outcomes & Future Work

This is where you finish the story.

Start:

“Coming to the current status of my project, I have completed the initial literature study, understood the sparse recovery problem and implemented the basic measurement model.”

Then:

“I have also studied and implemented the GAN-based reconstruction approach and performed initial experiments using MNIST.”

Then:

“For comparison, I have implemented LASSO and SBL and started analysing their reconstruction performance using MSE, NMSE and NMSE in dB.”

Learning

Then say:

“Through this work, I have learned both the theoretical and practical aspects of the problem. The main areas I learned are sparse signal recovery, measurement models, DNN and GAN concepts, LASSO, SBL, mathematical error evaluation and Python-based implementation.”

Future work

Finally:

“As the next step, I plan to test the approach on additional datasets, experiment with different measurement dimensions, further tune the GAN architecture and training parameters, and perform a more extensive comparison with conventional reconstruction methods.”

Finish with:

“These are the current learning outcomes and planned next steps of my project. Thank you.”

mathematical concepts you should study and be able to explain during your review/viva.

1. Vector and Matrix

An image can be represented as a vector.

For MNIST:

$$28 \times 28 = 784$$

So the image becomes:

$$x \in \mathbb{R}^{784}$$

A matrix is used to take measurements:

$$A \in \mathbb{R}^{196 \times 784}$$

2. Measurement Model

This is the most important equation in your project:

$$\boxed{y = Ax}$$

Where:

- x = original image
- A = measurement/compression matrix
- y = measured/low-dimensional data

In your experiment:

$$784 \rightarrow 196$$

So,

$$x(784) \xrightarrow{A} y(196)$$

The reconstruction task is:

$$y \rightarrow \hat{x}$$

3. Underdetermined System

You have:

$$M < N$$

In your case:

$$196 < 784$$

There are fewer measurements than unknown pixels.

Therefore, directly solving

$$Ax = y$$

does not normally give a unique solution.

This is why **sparse recovery and learned reconstruction** are required.

4. Linear Algebra

You should know:

- Matrix multiplication
- Transpose
- Inverse
- Rank
- Norm
- Dot product
- Eigenvalues/eigenvectors — basic understanding is enough

For example:

$$y = Ax$$

is a matrix multiplication operation.

5. Norms

A norm measures the **size or magnitude of a vector**.

The commonly used one in your project is the L_2 norm:

$$\|x\|_2 = \sqrt{\sum_i x_i^2}$$

The squared L_2 norm is:

$$\|x\|_2^2 = \sum_i x_i^2$$

You use this in NMSE.

6. L1 Norm

LASSO uses the L_1 norm:

$$\|x\|_1 = \sum_i |x_i|$$

LASSO solves approximately:

$$\hat{x} = \arg \min_x \left[\frac{1}{2} \|Ax - y\|_2^2 + \alpha \|x\|_1 \right]$$

The L_1 term encourages a **sparse solution**.

7. Sparse Signal

A signal is called sparse when most of its important coefficients are zero or close to zero.

For example:

$$x = [0, 0, 5, 0, 0, 2, 0, 0]$$

Only a few values contain significant information.

Sparse recovery tries to recover such information from fewer measurements.

8. MSE — Mean Squared Error

You compare the original image x and reconstructed image $\hat{x}$:

$$MSE = \frac{1}{N} \sum_{i=1}^N (x_i - \hat{x}_i)^2$$

Simple meaning:

MSE tells how much the reconstructed pixels differ from the original pixels.

Lower MSE generally means smaller reconstruction error.

9. NMSE — Normalized Mean Squared Error

Your project also uses normalized error:

$$NMSE = \frac{\|x - \hat{x}\|_2^2}{\|x\|_2^2}$$

Why normalize?

Because it makes the error relative to the energy/magnitude of the original signal.

10. NMSE in dB

You convert NMSE into decibels:

$$NMSE_{dB} = 10 \log_{10}(NMSE)$$

For example, if:

$$NMSE = 0.1$$

then:

$$10 \log_{10}(0.1) = -10 \text{ dB}$$

So your guide may ask:

“Why is your NMSE(dB) negative?”

Answer:

Because the NMSE is less than 1. When a value between 0 and 1 is converted using $10\log_{10}$, the result becomes negative.

11. GAN Mathematical Concept

Your generator performs:

$$\hat{x} = G(y)$$

Here:

- y = 196-dimensional measurement
- G = Generator
- $\hat{x}$ = reconstructed 784-dimensional image

Your project is therefore:

$$y = Ax \rightarrow G(y) \rightarrow \hat{x}$$

12. Loss Function

The GAN training process uses a loss function to measure how well the generated image is performing.

For reconstruction, MSE can be used:

$$L_{rec} = \frac{1}{N} \sum_{i=1}^N (x_i - \hat{x}_i)^2$$

The model updates its weights to reduce the loss.

Basic optimization idea:

$$\theta_{new} = \theta_{old} - \eta \nabla_{\theta} L$$

where:

- θ = model weights
- η = learning rate

- L = loss
 - ∇L = gradient
-

13. Gradient Descent

This is the basic mathematical idea behind neural-network training.

The model:

1. Produces reconstruction.
2. Calculates error.
3. Calculates gradient.
4. Updates weights.
5. Repeats.

Conceptually:

Weights $\rightarrow$ Prediction $\rightarrow$ Loss $\rightarrow$ Gradient $\rightarrow$ Updated Weights

14. SBL Mathematical Concept

Sparse Bayesian Learning (SBL) treats the unknown signal x probabilistically.

A basic idea is:

$$x_i \sim \mathcal{N}(0, \gamma_i)$$

where γ_i controls how important each coefficient is.

SBL iteratively estimates these parameters and uses them to obtain a sparse reconstruction.

You don't need to memorize the complete SBL derivation for your presentation unless your guide specifically asks for it.

What you should study first

For your project, prioritize these:

Priority	Mathematical concept
★★★★★	$y = Ax$
★★★★★	Matrix multiplication
★★★★★	$M < N$, underdetermined system
★★★★★	MSE
★★★★★	NMSE
★★★★★	NMSE(dB)
★★★★	L_1 and L_2 norms
★★★★	Sparse signal
★★★★	LASSO equation
★★★★	Gradient descent
★★★	GAN loss
★★★	SBL basic mathematics
★★	Eigenvalues/eigenvectors
★★	Rank

The most important chain to understand for your viva is:

$$xy = Ax\hat{x}$$

followed by

$$x, \hat{x} \rightarrow MSE \rightarrow NMSE \rightarrow NMSE(dB)$$

If you understand this chain clearly, you can explain most of the mathematics of your current project.

-Minute Presentation Script

Slide 1 — Title

Time: 20 seconds

Good morning everyone.

I am Aravind Ashok from the Department of Computer Science and Engineering.

Today I am presenting the progress of my M.Tech project titled **“High-Resolution Image Reconstruction Using GAN.”**

In this presentation, I will explain the problem, the mathematical model, the GAN-based reconstruction approach, the conventional methods I implemented, and the current results.

Slide 2 — Introduction

Time: 40 seconds

An image contains a large number of pixels, so it can be represented as a high-dimensional signal.

Instead of using the complete image directly, we take a smaller number of measurements and try to reconstruct the original image from those measurements.

The basic measurement model is:

$$\begin{bmatrix} y = Ax \end{bmatrix}$$

Here, (x) is the original image, (A) is the measurement matrix, and (y) is the low-dimensional measurement.

The main objective of my project is to reconstruct the image from this low-dimensional measurement using a GAN.

Slide 3 — Problem Statement

Time: 40 seconds

For MNIST, each image has a size of (28×28) , so there are 784 pixels.

Therefore,

$$\begin{bmatrix} N = 784 \end{bmatrix}$$

In my experiment, I use 196 measurements:

[
 $M=196$
]

So,

[
 $M < N$
]

This creates an underdetermined reconstruction problem because we have fewer measurements than the original number of pixels.

The objective is to recover an image ($\hat{x}$) that is as close as possible to the original image (x).

Slide 4 — Reference Paper

Time: 40 seconds

The main reference paper for my work is “**Sparse Signal Recovery Using Deep Neural Networks**” by Rubin Jose Peter and Chandra R. Murthy.

This paper helped me understand the basic concept of recovering a signal from limited measurements.

From this paper, I studied the measurement model, sparse recovery, reconstruction from fewer measurements, and reconstruction error.

My current work extends this general reconstruction idea by investigating a **GAN-based reconstruction approach** and comparing it with conventional methods such as LASSO and SBL.

Slide 5 — Mathematical Concepts

Time: 50 seconds

The main mathematical model used in my project is:

[
 $y = Ax$
]

The original image is represented as a vector:

$$[x \in \mathbb{R}^{784}]$$

The measurement matrix is:

$$[A \in \mathbb{R}^{196 \times 784}]$$

and the measurement vector is:

$$[y \in \mathbb{R}^{196}]$$

So the 784-dimensional image is converted into 196 measurements.

For evaluating reconstruction, I use MSE:

$$[MSE = \frac{1}{N} \sum_{i=1}^N (x_i - \hat{x}_i)^2]$$

I also use NMSE:

$$[NMSE = \frac{\|x - \hat{x}\|_2^2}{\|x\|_2^2}]$$

and convert it into decibels using:

$$[NMSE_{dB} = 10 \log_{10}(NMSE)]$$

These metrics measure the reconstruction error.

Slide 6 — Sparse Signal Recovery

Time: 35 seconds

Sparse signal recovery is the process of recovering a signal from a limited number of measurements by using the assumption that the signal has a sparse representation.

The important issue in my project is that:

[
 $M < N$
]

So direct recovery is difficult.

This motivated me to study conventional sparse recovery methods such as **LASSO** and **SBL**, and also learning-based reconstruction using GAN.

Slide 7 — DNN and GAN Learning

Time: 45 seconds

Before implementing the model, I studied the basic concepts of neural networks and GANs.

A GAN mainly contains two networks: a **Generator** and a **Discriminator**.

The Generator produces an image, while the Discriminator tries to distinguish between real and generated images.

In my reconstruction problem, instead of using only random noise as the Generator input, I use the measurement vector (y).

So my reconstruction path is:

[
 $y \rightarrow G(y) \rightarrow \hat{x}$
]

where (G) is the Generator and ($\hat{x}$) is the reconstructed image.

Slide 8 — Proposed GAN Method

Time: 45 seconds

The overall methodology is:

First, I take the MNIST image.

The image is normalized and flattened from (28×28) into a 784-dimensional vector.

Then I multiply it with the measurement matrix:

[
 $y = Ax$
]

This produces the 196-dimensional measurement.

The measurement is given to the Generator.

The Generator produces a 784-dimensional reconstructed image:

```
[  
  \hat{x}=G(y)  
]
```

Finally, I reshape the output back into a (28×28) image and evaluate its reconstruction error.

Slide 9 — GAN Architecture

Time: 40 seconds

The main input to my Generator is the 196-dimensional measurement.

The Generator consists of fully connected layers and produces 784 output values.

These 784 values represent the reconstructed image.

During GAN training, the Discriminator receives real and generated images and helps the Generator learn to produce more realistic reconstructions.

During reconstruction, the important path is:

```
[  
  196 \rightarrow \text{Generator} \rightarrow 784  
]
```

and finally:

```
[  
  784 \rightarrow 28 \times 28  
]
```

Slide 10 — Software and Implementation

Time: 35 seconds

I implemented the project using Python.

The main tools and libraries are:

- **NumPy** for arrays and matrix operations

- **TensorFlow/Keras** for GAN implementation
- **Matplotlib** for graphs and reconstructed images
- **Scikit-learn** for LASSO
- **SciPy** for numerical operations
- **Struct** for reading the MNIST IDX files

I used both VS Code and Google Colab during development and testing.

Slide 11 — Dataset and Experimental Setup

Time: 35 seconds

For the initial experiments, I used the **MNIST handwritten digit dataset**.

Each image is:

```
[  
28\times28  
]
```

or 784 pixels.

My main experiment uses:

```
[  
N=784  
]
```

```
[  
M=196  
]
```

and 10,000 training samples.

I also experimented with reducing the number of measurements to observe how measurement reduction affects reconstruction quality.

Slide 12 — GAN Results

Time: 45 seconds

Here I show the original and reconstructed MNIST images.

The reconstructed images are able to preserve the main structure of the digits.

I observed that the number of measurements has a significant effect on reconstruction quality.

When the number of measurements is reduced, more information is lost, which makes reconstruction more difficult and increases the reconstruction error.

I evaluated the results using MSE, NMSE and NMSE in dB, along with visual comparison.

Slide 13 — LASSO and SBL

Time: 45 seconds

To compare the proposed approach with conventional reconstruction techniques, I implemented **LASSO and SBL**.

LASSO uses (L_1) regularization:

$$\begin{aligned} [\hat{x} = & \\ \arg\min_x & \\ \left[\right. & \\ \frac{1}{2} \|Ax - y\|_2^2 + \alpha \|x\|_1 & \\ \left. \right] & \end{aligned}$$

It encourages a sparse solution.

SBL, or Sparse Bayesian Learning, uses a Bayesian approach and iteratively estimates the importance of the signal coefficients.

Both methods use the same measurement setup so that the comparison is consistent.

Slide 14 — Comparison

Time: 50 seconds

Here I compare the three approaches:

GAN, LASSO and SBL.

For a fair comparison, the same type of input image and measurement process is used.

The output of each method is a reconstructed image.

I compare them using:

- MSE

- NMSE
- NMSE(dB)
- Visual reconstruction quality

The graph shows the reconstruction performance of the three methods based on the values obtained from my experiments.

This comparison helps me understand the difference between a learning-based reconstruction approach and conventional sparse recovery methods.

Slide 15 — Current Status and Future Work

Time: 40 seconds

At the current stage, I have completed the literature study and learned the mathematical basis of sparse signal recovery.

I have implemented the measurement model, GAN-based reconstruction, LASSO and SBL.

I have also implemented MSE, NMSE and NMSE in dB for evaluating the reconstruction.

As future work, I plan to improve the GAN reconstruction quality, experiment with different measurement ratios, tune the network architecture, test additional datasets, and perform a more detailed comparison.

For your **10-minute M.Tech project presentation on High-Resolution Image Reconstruction Using GAN**, your guide/examiners will most likely ask questions from **four areas: basic concept, mathematics, GAN, and your implementation/results**.

Here are the questions you should prepare.

1. Basic project questions — very likely

Q1. What is your project about?

Answer:

My project is about reconstructing an image from a smaller number of measurements using a GAN. I use MNIST images, convert the 784-pixel image into 196 measurements using $y = Ax$, and use a GAN-based generator to reconstruct the 784-pixel image.

Q2. What is the main problem you are solving?

The main problem is image reconstruction from limited measurements. Since I use 196 measurements for a 784-pixel image, information is reduced and the original image has to be reconstructed from incomplete information.

Q3. Why do you need image reconstruction?

Reconstruction is useful when storing or transmitting the complete image is expensive. Instead, fewer measurements can be acquired and the original information can be approximately recovered.

Q4. What is sparse signal recovery?

Sparse signal recovery is the process of recovering a signal from a limited number of measurements by exploiting sparsity or other prior information about the signal.

2. Mathematical questions — VERY important

Q5. What is $y = Ax$?

x is the original signal or image, A is the measurement matrix, and y is the compressed or low-dimensional measurement.

$$y = Ax$$

In my experiment:

$$x = 784$$

and

$$y = 196$$

Q6. Why is $M < N$ important?

$M < N$ means that the number of measurements is smaller than the number of unknown values. Therefore, the system is underdetermined and direct recovery is difficult.

For my experiment:

$$196 < 784$$

Q7. What is an underdetermined system?

It is a system where there are fewer equations than unknown variables. Therefore, there can be multiple possible solutions, and additional assumptions or methods are required to find a suitable solution.

Q8. What is the dimension of your measurement matrix?

For $N = 784$ and $M = 196$:

$$A \in \mathbb{R}^{196 \times 784}$$

Because:

$$(196 \times 784)(784 \times 1) = 196 \times 1$$

Q9. What is MSE?

MSE means Mean Squared Error. It measures the average squared difference between the original and reconstructed image.

$$MSE = \frac{1}{N} \sum_{i=1}^N (x_i - \hat{x}_i)^2$$

Q10. What is NMSE?

NMSE is Normalized Mean Squared Error. It normalizes the reconstruction error with respect to the energy of the original signal.

$$NMSE = \frac{\|x - \hat{x}\|_2^2}{\|x\|_2^2}$$

Q11. Why do you use NMSE instead of only MSE?

MSE gives the absolute error, while NMSE gives the error relative to the original signal. Therefore, NMSE is useful for comparing reconstruction performance more consistently.

Q12. Why is your NMSE(dB) negative?

This is a **very likely question**.

Because the NMSE is less than 1. When we calculate $10\log_{10}(NMSE)$, a value between 0 and 1 produces a negative dB value.

For example:

$$NMSE = 0.1$$
$$10\log_{10}(0.1) = -10 \text{ dB}$$

3. GAN questions

Q13. What is GAN?

GAN stands for Generative Adversarial Network. It consists mainly of a Generator and a Discriminator. The Generator produces an output, and the Discriminator tries to distinguish real images from generated images.

Q14. What is the Generator doing in your project?

The Generator takes the 196-dimensional measurement vector as input and reconstructs a 784-dimensional image.

$$\hat{x} = G(y)$$

Q15. What is the input to your Generator?

This is important because your project is different from a standard GAN.

The input is the measurement vector y , containing 196 measurements. I use the measurement vector instead of using only a random noise vector as the reconstruction input.

Q16. What is the output of your Generator?

The Generator produces 784 values, which are reshaped into a 28×28 reconstructed MNIST image.

So:

$$196 \rightarrow 784 \rightarrow 28 \times 28$$

Q17. What is the role of the Discriminator?

The Discriminator tries to distinguish between real images and reconstructed/generated images. Its feedback helps the Generator produce more realistic images during GAN training.

Q18. Why did you use GAN?

A safe answer:

I selected GAN because it provides a learning-based approach for generating realistic image structures from limited information. My project investigates whether this approach can be used for reconstruction from sparse measurements.

Don't say “**GAN is always better**”. Your experimental results should establish that for your particular setup.

4. LASSO questions

Q19. What is LASSO?

LASSO stands for Least Absolute Shrinkage and Selection Operator. It is a sparse recovery method that uses L_1 regularization.

$$\hat{x} = \arg \min_x \left[\frac{1}{2} \|Ax - y\|_2^2 + \alpha \|x\|_1 \right]$$

Q20. Why does LASSO use L_1 norm?

The L_1 regularization encourages many coefficients to become zero or close to zero, which promotes sparsity.

Q21. What is α in LASSO?

α controls the strength of the L_1 regularization. A larger value puts more emphasis on sparsity, while a smaller value puts more emphasis on fitting the measurements.

5. SBL questions

Q22. What is SBL?

SBL stands for Sparse Bayesian Learning. It is a Bayesian sparse recovery method that estimates the unknown signal using probabilistic modeling and iteratively estimates parameters related to the importance of different coefficients.

Q23. How is SBL different from LASSO?

Simple answer:

LASSO uses L_1 regularization to promote sparsity, while SBL uses a Bayesian probabilistic framework to obtain a sparse solution.

6. Your experiment questions

Q24. Why did you use MNIST?

MNIST is a standard image dataset with simple 28×28 grayscale images. It is suitable for initially testing reconstruction algorithms because the data size and structure are manageable.

Q25. Why 28×28 ?

MNIST images are provided as 28×28 grayscale images. Therefore, flattening gives:

$$28 \times 28 = 784$$

Q26. Why did you choose $M = 196$?

I selected 196 measurements so that the measurement dimension is significantly smaller than 784 while still retaining enough information to demonstrate reconstruction.

You can also mention:

$$\frac{196}{784} = 25\%$$

So you are using **25% of the original dimensionality as measurements**.

Q27. What happens if you reduce M further?

Less information is available for reconstruction, so the reconstruction becomes more difficult and the image quality can decrease. In my experiments, reducing the measurement dimension resulted in higher reconstruction error.

Q28. How did you evaluate your model?

I used MSE, NMSE, NMSE(dB), and visual comparison between original and reconstructed images.

7. Implementation questions

Q29. Which programming language did you use?

Python.

Q30. Which libraries did you use?

NumPy, TensorFlow/Keras, Matplotlib, Scikit-learn, SciPy, and Struct.

Q31. What is NumPy used for?

Matrix operations, arrays, numerical calculations, and generating the measurement matrix.

Q32. What is TensorFlow/Keras used for?

Building and training the GAN model.

Q33. What is Matplotlib used for?

Plotting loss curves, comparison graphs, and displaying reconstructed images.

Q34. Why did you use Scikit-learn?

I used Scikit-learn to implement LASSO for conventional comparison.

8. Results questions

Q35. What did you observe from your experiments?

I observed that reconstruction quality depends strongly on the number of measurements. With more measurements, more information is available for reconstruction. When measurements are reduced, reconstruction becomes more difficult.

Q36. Why are some reconstructed images blurry?

Because the reconstruction is performed from limited measurements. Some information about the original image is lost during measurement, so the reconstruction method has to estimate the missing information.

Q37. Which method gives better results?

For this question, **show your actual measured values** rather than answering from expectation.

Say:

“According to my current experimental results, the measured MSE/NMSE values are shown here. The visual and numerical results need to be considered together.”

Do not claim a general winner based on only one experiment.

9. Questions they may ask to test whether you actually understand the code

These are particularly important.

Q38. What does this line do?

```
y = x @ A.T
```

You should understand that it performs the measurement operation corresponding to:

$$y = Ax$$

depending on how A is stored/transposed in your code.

Q39. Why do you flatten the image?

Because matrix-based measurement requires the image to be represented as a vector. A 28×28 image is converted into a 784-dimensional vector.

Q40. Why do you normalize the MNIST images?

Pixel values are originally in the range 0 to 255. Normalization converts them to a smaller numerical range, typically 0 to 1, which makes neural-network training more stable.

10. The hardest questions you should prepare

Your guide may ask these:

Q41. If you have only 196 measurements, how can you recover 784 values?

Answer:

Directly, it is an underdetermined problem. The reconstruction is possible approximately by using prior information about the image and a learned mapping. In my project, the Generator learns a mapping from the 196-dimensional measurement to the 784-dimensional reconstructed image.

Q42. What information is lost during measurement?

Since 784 values are represented using only 196 measurements, some information about the original image is not directly available. The reconstruction method estimates the missing information using the learned or assumed structure of the data.

Q43. What is the main difference between your GAN and LASSO?

GAN learns a nonlinear mapping from measurements to reconstructed images using training data, whereas LASSO solves an optimization problem using L_1 regularization to obtain a sparse solution.

Q44. What is the main difference between GAN and SBL?

GAN is a learning-based reconstruction approach that learns the mapping from measurements to images. SBL is a Bayesian sparse recovery approach that estimates the signal and its associated parameters iteratively.

Q45. What is the novelty of your project?

Be careful here. Don't claim that GAN-based reconstruction itself is universally novel.

A safe answer is:

“The project investigates a GAN-based reconstruction formulation where the sparse measurement vector is used as the Generator input, and evaluates it alongside LASSO and SBL under the same measurement setup. The specific novelty claim will depend on the literature review and the final experimental design.”

★ Memorize these 10 first

If you have limited preparation time, prepare these extremely well:

1. **What is your project?**
2. **What is $y = Ax$?**
3. **Why $M < N$?**
4. **What is sparse signal recovery?**
5. **Why GAN?**

6. **What is the Generator doing?**
7. **What is the input/output dimension?**
8. **What are MSE, NMSE and NMSE(dB)?**
9. **Why LASSO and SBL?**
10. **What did you actually observe from your experiments?**

If you can answer these naturally without reading from the PPT, you will be much better prepared for the 10-minute review.

C

Yes. For your **10-minute M.Tech review**, you should clearly show **what you learned**, not only what you coded. This is especially important because your project is still in progress.

I would restructure the presentation so that your learning is visible throughout the presentation rather than putting everything under one “Learning” slide.

Recommended 10-minute structure

Slide	Topic	Time
1	Title	15 sec
2	Project Introduction	35 sec
3	Problem + Objective	40 sec
4	Reference Paper & Literature Learning	45 sec
5	Mathematical Concepts Learned	1 min
6	Sparse Recovery: LASSO & SBL	1 min
7	GAN Concepts Learned	1 min
8	Proposed Method & Implementation	1 min
9	Results & Comparison	1.5 min
10	Learning Outcomes + Current Status + Future Work	1 min
Total		~9–10 min

What to show as your "learning"

1. Literature learning

Don't just say:

“I studied some papers.”

Show what you learned from them:

Literature Study

- Sparse signal recovery
- Reconstruction from limited measurements
- Measurement model $y = Ax$
- Relationship between measurements and signal dimension
- Deep-learning-based reconstruction
- Reconstruction error evaluation

Then explain:

“The reference paper helped me understand the basic sparse recovery problem. Based on that foundation, I studied GAN-based reconstruction for my project.”

2. Mathematical learning

This should be one of your strongest slides.

Put:

$$\begin{aligned}y &= Ax \\x &\in R^{784} \\A &\in R^{196 \times 784} \\y &\in R^{196}\end{aligned}$$

Then:

Concepts Learned

- Vectors and matrices
- Matrix multiplication
- Underdetermined systems
- L_1 norm
- L_2 norm
- Sparse representation

- MSE
- NMSE
- NMSE(dB)

When presenting, say:

“During the project, I learned how an image can be represented mathematically as a vector and how a measurement matrix converts the 784-dimensional image into 196 measurements. I also learned how reconstruction error is measured using MSE, NMSE and NMSE in dB.”

That directly demonstrates learning.

3. LASSO and SBL learning

Show that you didn't simply run code.

What I learned

LASSO

- L_1 regularization
- Sparse solution
- Optimization problem
- Regularization parameter α

SBL

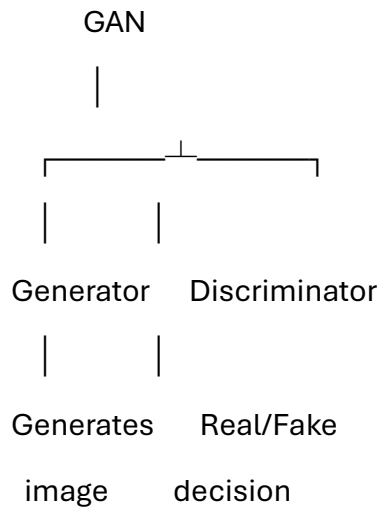
- Bayesian approach
- Sparse recovery
- Hyperparameter estimation
- Iterative reconstruction

Then say:

“I studied these two conventional sparse recovery approaches so that I could compare them with my GAN-based reconstruction.”

4. GAN learning

Show the standard concept first:



Then show **your adaptation**:

$$y \rightarrow G(y) \rightarrow \hat{x}$$

Explain:

“I first learned the basic GAN architecture of Generator and Discriminator. Then I adapted the concept to my reconstruction problem by using the measurement vector as the Generator input.”

This is a good way to demonstrate progression from **learning** → **implementation**.

5. Programming/technical learning

You should also show what you learned practically.

Technical Learning

- Python for numerical computation
- NumPy for matrices and arrays
- TensorFlow/Keras for GAN
- Scikit-learn for LASSO
- SciPy for numerical operations
- Matplotlib for visualization
- MNIST IDX file handling
- Image normalization
- Image flattening

- Matrix-based measurement generation
- Model training
- Performance evaluation

You don't need to explain every library. Just mention the important ones.

Your final slide should emphasize learning

I recommend changing your final slide to:

Learning Outcomes & Current Status

Learning

- Understood sparse signal recovery
- Learned mathematical model $y = Ax$
- Learned underdetermined systems
- Learned L_1/L_2 norms
- Learned MSE, NMSE and NMSE(dB)
- Learned GAN architecture and training
- Learned LASSO and SBL

Implementation

- MNIST preprocessing
- Measurement generation
- GAN reconstruction
- LASSO reconstruction
- SBL reconstruction
- Performance comparison

Current Status

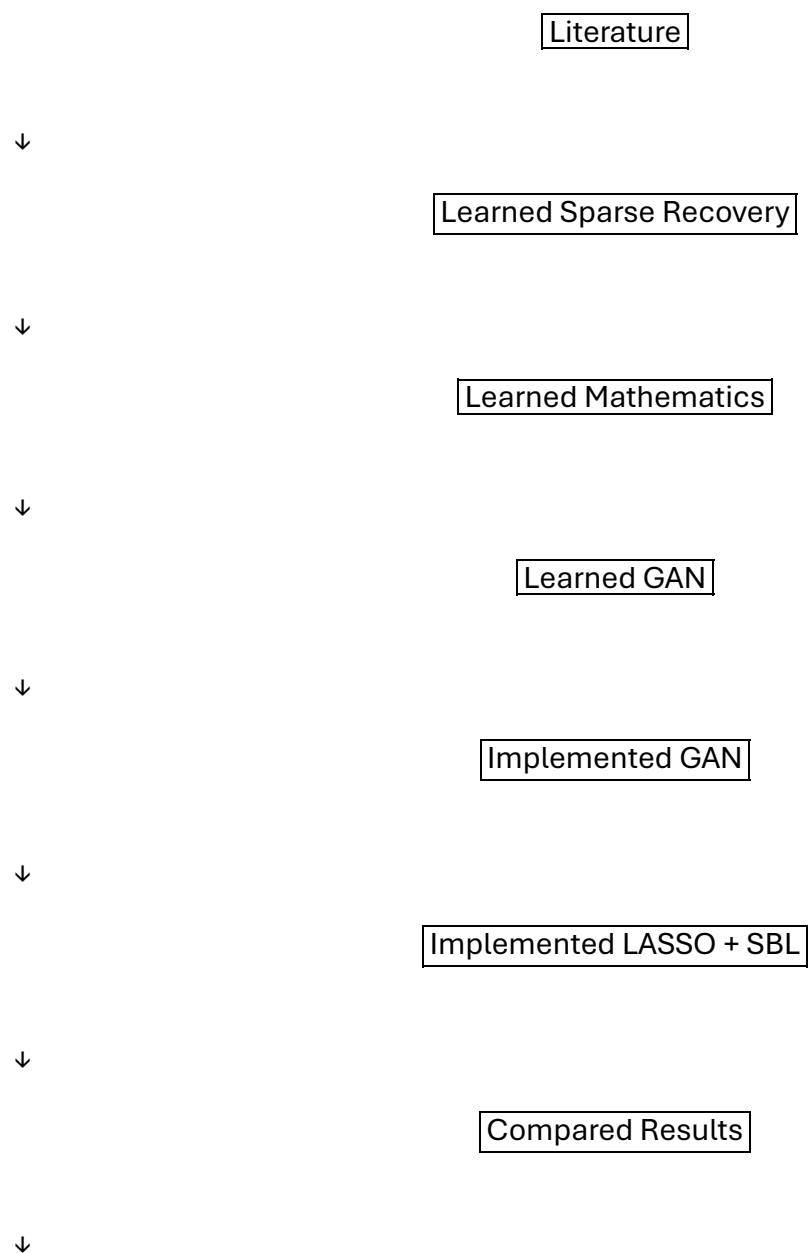
- GAN reconstruction implemented
- Conventional methods implemented
- Initial comparison completed

Future Work

- Improve reconstruction quality
 - Test different measurement ratios
 - Tune GAN architecture
 - Test additional datasets
 - Complete detailed evaluation
-

The important story you should tell

Your presentation should make the examiner see this progression:



Next: Improve and Extend

That is much stronger for a **project-progress/learning review** than presenting it as if the entire project is already finished.